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Roll no: 24

Subject: Enterprise Java Technology

Assignment – 1

Q-1. Write a Java program to establish a connection between a Java application and a database using JDBC and display an appropriate message after successful connection.

```
import java.sql.Connection;
import java.sql.DriverManager;
public class JDBCConnection {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                    url, user, password);
            System.out.println("Database connected successfully!");
            con.close();
        }
        catch (Exception e) {
            System.out.println("Connection failed!");
            System.out.println(e);
        }
    }
}
```

Q-2. Write a Java program using JDBC to create a Student table containing Student ID, Student Name, Course, Semester, and Marks.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;
public class CreateStudentTable {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                    url, user, password);
            Statement stmt = con.createStatement();
            String sql = "CREATE TABLE IF NOT EXISTS student ("
                    + "student_id INT PRIMARY KEY, "
                    + "student_name VARCHAR(50), "
                    + "course VARCHAR(50), "
                    + "semester INT, "
                    + "marks DOUBLE)";
            stmt.executeUpdate(sql);
            System.out.println("Student table created successfully!");
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
```

```
}
```

Q-3. Write a Java program using JDBC to insert student details into the Student table.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;
import java.util.Scanner;
public class InsertStudent {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        Scanner sc = new Scanner(System.in);
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                    url, user, password);
            Statement stmt = con.createStatement();
            System.out.print("Enter Student ID: ");
            int id = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter Student Name: ");
            String name = sc.nextLine();
            System.out.print("Enter Course: ");
            String course = sc.nextLine();
            System.out.print("Enter Semester: ");
            int semester = sc.nextInt();
            System.out.print("Enter Marks: ");
```

```
            double marks = sc.nextDouble();
            String sql = "INSERT INTO student VALUES ("
                    + id + ", '"
                    + name + "', '"
                    + course + "', "
                    + semester + ", "
                    + marks + ")";
            stmt.executeUpdate(sql);
            System.out.println("Student inserted successfully!");
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Q-4. Write a Java program using JDBC to retrieve and display all records from the Student table using ResultSet.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
public class DisplayStudent {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            Connection con = DriverManager.getConnection(
                    url, user, password);
            Statement stmt = con.createStatement();
            String sql = "SELECT * FROM student";
            ResultSet rs = stmt.executeQuery(sql);
            System.out.println("ID\tName\tCourse\tSemester\tMarks");
            System.out.println("-----------------------------------------------");
            while (rs.next()) {
                System.out.println(
                        rs.getInt("student_id") + "\t" +
                        rs.getString("student_name") + "\t" +
                        rs.getString("course") + "\t" +
                        rs.getInt("semester") + "\t\t" +
                        rs.getDouble("marks")
                );
            }
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Q-5. Write a Java program using JDBC to search for a student record based on the Student ID entered by the user.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
```

```
import java.sql.Statement;
import java.util.Scanner;
public class SearchStudent {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        Scanner sc = new Scanner(System.in);
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();
            System.out.print("Enter Student ID to search: ");
            int id = sc.nextInt();
            String sql = "SELECT * FROM student WHERE student_id = " + id;
            ResultSet rs = stmt.executeQuery(sql);
            if (rs.next()) {
                System.out.println("Student Found!");
                System.out.println("Student ID: " + rs.getInt("student_id"));
                System.out.println("Name: " + rs.getString("student_name"));
                System.out.println("Course: " + rs.getString("course"));
                System.out.println("Semester: " + rs.getInt("semester"));
                System.out.println("Marks: " + rs.getDouble("marks"));
            }
            else {
                System.out.println("Student not found!");
            }
            con.close();
        }
```

```
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Q-6. Write a Java program using JDBC to update the marks of a student based on the Student ID.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;
import java.util.Scanner;
public class UpdateMarks {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        Scanner sc = new Scanner(System.in);
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();
            System.out.print("Enter Student ID: ");
            int id = sc.nextInt();
            System.out.print("Enter New Marks: ");
            double marks = sc.nextDouble();
            String sql = "UPDATE student SET marks = " + marks + " WHERE student_id
= " + id;
            int rows = stmt.executeUpdate(sql);
            if (rows > 0) {
```

```
            System.out.println("Marks updated successfully!");
        }
        else {
            System.out.println("Student not found!");
        }
        con.close();
    }
    catch (Exception e) {
        System.out.println(e);
    }
  }
}
```

Q-7. Write a Java program using JDBC to delete a student record based on the Student ID.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;
import java.util.Scanner;
public class DeleteStudent {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        Scanner sc = new Scanner(System.in);
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();
```

```
            System.out.print("Enter Student ID to delete: ");
            int id = sc.nextInt();
            String sql = "DELETE FROM student WHERE student_id = " + id;
            int rows = stmt.executeUpdate(sql);
            if (rows > 0) {
                System.out.println("Student deleted successfully!");
            }
            else {
                System.out.println("Student not found!");
            }
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Q-8. Write a Java program using PreparedStatement to insert student details into a database table.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.util.Scanner;
public class PreparedInsert {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
```

```
        Scanner sc = new Scanner(System.in);
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            String sql = "INSERT INTO student (student_id, student_name, course,
semester, marks) VALUES (?, ?, ?, ?, ?)";
            PreparedStatement ps = con.prepareStatement(sql);
            System.out.print("Enter Student ID: ");
            int id = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter Student Name: ");
            String name = sc.nextLine();
            System.out.print("Enter Course: ");
            String course = sc.nextLine();
            System.out.print("Enter Semester: ");
            int semester = sc.nextInt();
            System.out.print("Enter Marks: ");
            double marks = sc.nextDouble();
            ps.setInt(1, id);
            ps.setString(2, name);
            ps.setString(3, course);
            ps.setInt(4, semester);
            ps.setDouble(5, marks);
            ps.executeUpdate();
            System.out.println("Student inserted successfully!");
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
```

```
        }
    }
}
```

Q-9. Write a Java program using PreparedStatement to search for a student record using Student ID.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;
public class PreparedSearch {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        Scanner sc = new Scanner(System.in);
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            String sql = "SELECT * FROM student WHERE student_id = ?";
            PreparedStatement ps = con.prepareStatement(sql);
            System.out.print("Enter Student ID: ");
            int id = sc.nextInt();
            ps.setInt(1, id);
            ResultSet rs = ps.executeQuery();
            if (rs.next()) {
                System.out.println("Student Found!");
                System.out.println("Student ID: " + rs.getInt("student_id"));
```

```
            System.out.println("Name: " + rs.getString("student_name"));
            System.out.println("Course: " + rs.getString("course"));
            System.out.println("Semester: " + rs.getInt("semester"));
            System.out.println("Marks: " + rs.getDouble("marks"));
        }
        else {
            System.out.println("Student not found!");
        }
        con.close();
    }
    catch (Exception e) {
        System.out.println(e);
    }
  }
}
```

Q-10. Write a menu-driven Java JDBC application to perform insertion, display, searching, updating, and deletion of student records.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;
public class StudentManagement {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        Scanner sc = new Scanner(System.in);
```

```
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            int choice;
            do {
                System.out.println("\n===== STUDENT MANAGEMENT =====");
                System.out.println("1. Insert Student");
                System.out.println("2. Display Students");
                System.out.println("3. Search Student");
                System.out.println("4. Update Marks");
                System.out.println("5. Delete Student");
                System.out.println("6. Exit");
                System.out.print("Enter your choice: ");
                choice = sc.nextInt();
                switch (choice) {
                    case 1:
                        String insert = "INSERT INTO student (student_id, student_name,
course, semester, marks) VALUES (?, ?, ?, ?, ?)";
                        PreparedStatement ps1 = con.prepareStatement(insert);
                        System.out.print("Enter Student ID: ");
                        int id = sc.nextInt();
                        sc.nextLine();
                        System.out.print("Enter Student Name: ");
                        String name = sc.nextLine();
                        System.out.print("Enter Course: ");
                        String course = sc.nextLine();
                        System.out.print("Enter Semester: ");
                        int semester = sc.nextInt();
                        System.out.print("Enter Marks: ");
```

```
                double marks = sc.nextDouble();
                ps1.setInt(1, id);
                ps1.setString(2, name);
                ps1.setString(3, course);
                ps1.setInt(4, semester);
                ps1.setDouble(5, marks);
                ps1.executeUpdate();
                System.out.println("Student inserted successfully!");
                break;
            case 2:
                String select = "SELECT * FROM student";
                PreparedStatement ps2 = con.prepareStatement(select);
                ResultSet rs = ps2.executeQuery();
                System.out.println("ID\tName\tCourse\tSemester\tMarks");
                System.out.println("-----------------------------------------------");
                while (rs.next()) {
                    System.out.println(rs.getInt("student_id") + "\t" +
rs.getString("student_name") + "\t" + rs.getString("course") + "\t" +
rs.getInt("semester") + "\t\t" + rs.getDouble("marks"));
                }
                break;
            case 3:
                String search = "SELECT * FROM student WHERE student_id = ?";
                PreparedStatement ps3 = con.prepareStatement(search);
                System.out.print("Enter Student ID: ");
                int searchId = sc.nextInt();
                ps3.setInt(1, searchId);
                ResultSet rs2 = ps3.executeQuery();
                if (rs2.next()) {
                    System.out.println("Student Found!");
```

```
                System.out.println("Student ID: " + rs2.getInt("student_id"));
                System.out.println("Name: " + rs2.getString("student_name"));
                System.out.println("Course: " + rs2.getString("course"));
                System.out.println("Semester: " + rs2.getInt("semester"));
                System.out.println("Marks: " + rs2.getDouble("marks"));
            }
            else {
                System.out.println("Student not found!");
            }
            break;
        case 4:
            String update = "UPDATE student SET marks = ? WHERE student_id
= ?";
            PreparedStatement ps4 = con.prepareStatement(update);
            System.out.print("Enter Student ID: ");
            int updateId = sc.nextInt();
            System.out.print("Enter New Marks: ");
            double newMarks = sc.nextDouble();
            ps4.setDouble(1, newMarks);
            ps4.setInt(2, updateId);
            int rows = ps4.executeUpdate();
            if (rows > 0) {
                System.out.println("Marks updated successfully!");
            }
            else {
                System.out.println("Student not found!");
            }
            break;
        case 5:
```

```
                String delete = "DELETE FROM student WHERE student_id = ?";
                PreparedStatement ps5 = con.prepareStatement(delete);
                System.out.print("Enter Student ID: ");
                int deleteId = sc.nextInt();
                ps5.setInt(1, deleteId);
                int deleted = ps5.executeUpdate();
                if (deleted > 0) {
                    System.out.println("Student deleted successfully!");
                }
                else {
                    System.out.println("Student not found!");
                }
                break;
            case 6:
                System.out.println("Program ended.");
                break;
            default:
                System.out.println("Invalid choice!");
          }
        } while (choice != 6);
        con.close();
    }
    catch (Exception e) {
        System.out.println(e);
    }
  }
}
```

Q-11. Write a Java JDBC program to maintain employee details such as Employee ID, Name, Department, Designation, and Salary and display the records.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.Statement;
import java.util.Scanner;
public class EmployeeDetails {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        Scanner sc = new Scanner(System.in);
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();
            String create = "CREATE TABLE IF NOT EXISTS employee (employee_id
INT PRIMARY KEY, name VARCHAR(50), department VARCHAR(50), designation
VARCHAR(50), salary DOUBLE)";
            stmt.executeUpdate(create);
            String insert = "INSERT INTO employee VALUES (?, ?, ?, ?, ?)";
            PreparedStatement ps = con.prepareStatement(insert);
            System.out.print("Enter Employee ID: ");
            int id = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter Name: ");
            String name = sc.nextLine();
```

```
            System.out.print("Enter Department: ");
            String department = sc.nextLine();
            System.out.print("Enter Designation: ");
            String designation = sc.nextLine();
            System.out.print("Enter Salary: ");
            double salary = sc.nextDouble();
            ps.setInt(1, id);
            ps.setString(2, name);
            ps.setString(3, department);
            ps.setString(4, designation);
            ps.setDouble(5, salary);
            ps.executeUpdate();
            System.out.println("Employee inserted successfully!");
            ResultSet rs = stmt.executeQuery("SELECT * FROM employee");
            System.out.println("\nID\tName\tDepartment\tDesignation\tSalary");
            System.out.println("-------------------------------------------------------------");
            while (rs.next()) {
                System.out.println(rs.getInt("employee_id") + "\t" + rs.getString("name") +
"\t" + rs.getString("department") + "\t" + rs.getString("designation") + "\t" +
rs.getDouble("salary"));
            }
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Q-12. Write a Java JDBC program to update the salary of an employee based on Employee ID.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.util.Scanner;
public class UpdateEmployeeSalary {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        Scanner sc = new Scanner(System.in);
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            String sql = "UPDATE employee SET salary = ? WHERE employee_id = ?";
            PreparedStatement ps = con.prepareStatement(sql);
            System.out.print("Enter Employee ID: ");
            int id = sc.nextInt();
            System.out.print("Enter New Salary: ");
            double salary = sc.nextDouble();
            ps.setDouble(1, salary);
            ps.setInt(2, id);
            int rows = ps.executeUpdate();
            if (rows > 0) {
                System.out.println("Salary updated successfully!");
            }
            else {
                System.out.println("Employee not found!");
            }
            con.close();
```

```
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Q-13. Write a Java JDBC program to maintain product details such as Product ID, Product Name, Category, Quantity, and Price.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.Statement;
import java.util.Scanner;
public class ProductDetails {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        Scanner sc = new Scanner(System.in);
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();
            String create = "CREATE TABLE IF NOT EXISTS product (product_id INT
PRIMARY KEY, product_name VARCHAR(50), category VARCHAR(50), quantity
INT, price DOUBLE)";
            stmt.executeUpdate(create);
            String insert = "INSERT INTO product VALUES (?, ?, ?, ?, ?)";
            PreparedStatement ps = con.prepareStatement(insert);
```

```
            System.out.print("Enter Product ID: ");
            int id = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter Product Name: ");
            String name = sc.nextLine();
            System.out.print("Enter Category: ");
            String category = sc.nextLine();
            System.out.print("Enter Quantity: ");
            int quantity = sc.nextInt();
            System.out.print("Enter Price: ");
            double price = sc.nextDouble();
            ps.setInt(1, id);
            ps.setString(2, name);
            ps.setString(3, category);
            ps.setInt(4, quantity);
            ps.setDouble(5, price);
            ps.executeUpdate();
            System.out.println("Product inserted successfully!");
            ResultSet rs = stmt.executeQuery("SELECT * FROM product");
            System.out.println("\nID\tName\tCategory\tQuantity\tPrice");
            System.out.println("-----------------------------------------------");
            while (rs.next()) {
                System.out.println(rs.getInt("product_id") + "\t" +
rs.getString("product_name") + "\t" + rs.getString("category") + "\t" +
rs.getInt("quantity") + "\t\t" + rs.getDouble("price"));
            }
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
```

```
        }
    }
}
```

Q-14. Write a Java JDBC program using ResultSetMetaData to display the number of columns and column names of a database table.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.ResultSetMetaData;
import java.sql.Statement;
public class ResultSetMetaDataDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery("SELECT * FROM student");
            ResultSetMetaData md = rs.getMetaData();
            int count = md.getColumnCount();
            System.out.println("Number of columns: " + count);
            System.out.println("Column names:");
            for (int i = 1; i <= count; i++) {
                System.out.println(md.getColumnName(i));
            }
            con.close();
        }
```

```
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Q-15. Write a Java JDBC program using DatabaseMetaData to display information about the connected database and JDBC driver.

```
import java.sql.Connection;
import java.sql.DatabaseMetaData;
import java.sql.DriverManager;
public class DatabaseMetaDataDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            DatabaseMetaData md = con.getMetaData();
            System.out.println("Database Product Name: " +
md.getDatabaseProductName());
            System.out.println("Database Version: " + md.getDatabaseProductVersion());
            System.out.println("JDBC Driver Name: " + md.getDriverName());
            System.out.println("JDBC Driver Version: " + md.getDriverVersion());
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
```

}

Q-16. Write a Java JDBC program to demonstrate transaction management using commit() and rollback().

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
public class TransactionDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        Connection con = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(url, user, password);
            con.setAutoCommit(false);
            String sql = "UPDATE student SET marks = ? WHERE student_id = ?";
            PreparedStatement ps = con.prepareStatement(sql);
            ps.setDouble(1, 90);
            ps.setInt(2, 101);
            ps.executeUpdate();
            ps.setDouble(1, 95);
            ps.setInt(2, 102);
            ps.executeUpdate();
            con.commit();
            System.out.println("Transaction committed successfully!");
            con.close();
        }
        catch (Exception e) {
```

```
            System.out.println("Transaction failed!");
            try {
                if (con != null) {
                    con.rollback();
                    System.out.println("Transaction rolled back!");
                }
            }
            catch (Exception ex) {
                System.out.println(ex);
            }
        }
    }
}
```

Q-17. Write a Java JDBC program to insert multiple records into a database using batch processing.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
public class BatchInsert {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            String sql = "INSERT INTO student VALUES (?, ?, ?, ?, ?)";
            PreparedStatement ps = con.prepareStatement(sql);
            ps.setInt(1, 201);
```

```
            ps.setString(2, "Amit");
            ps.setString(3, "MCA");
            ps.setInt(4, 3);
            ps.setDouble(5, 80);
            ps.addBatch();
            ps.setInt(1, 202);
            ps.setString(2, "Priya");
            ps.setString(3, "MCA");
            ps.setInt(4, 3);
            ps.setDouble(5, 88);
            ps.addBatch();
            ps.setInt(1, 203);
            ps.setString(2, "Neha");
            ps.setString(3, "MCA");
            ps.setInt(4, 3);
            ps.setDouble(5, 92);
            ps.addBatch();
            ps.executeBatch();
            System.out.println("Multiple records inserted successfully!");
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Q-18. Write a Java JDBC program to retrieve the maximum, minimum, average, and total salary from an Employee table using SQL aggregate functions.

```
import java.sql.Connection;
```

```
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
public class SalaryAggregate {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();
            String sql = "SELECT MAX(salary), MIN(salary), AVG(salary), SUM(salary)
FROM employee";
            ResultSet rs = stmt.executeQuery(sql);
            if (rs.next()) {
                System.out.println("Maximum Salary: " + rs.getDouble(1));
                System.out.println("Minimum Salary: " + rs.getDouble(2));
                System.out.println("Average Salary: " + rs.getDouble(3));
                System.out.println("Total Salary: " + rs.getDouble(4));
            }
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Q-19. Write a Java JDBC program to retrieve related information from two database tables using an SQL JOIN operation.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
public class JoinDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();
            String sql = "SELECT student.student_id, student.student_name,
course.course_name FROM student JOIN course ON student.course =
course.course_id";
            ResultSet rs = stmt.executeQuery(sql);
            System.out.println("ID\tName\tCourse");
            System.out.println("----------------------------");
            while (rs.next()) {
                System.out.println(rs.getInt("student_id") + "\t" +
rs.getString("student_name") + "\t" + rs.getString("course_name"));
            }
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Q-20. Write a Java JDBC program to calculate and display the total marks, percentage, and result of students using data retrieved from a database.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
public class StudentResult {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();
            String sql = "SELECT * FROM student";
            ResultSet rs = stmt.executeQuery(sql);
            while (rs.next()) {
                int id = rs.getInt("student_id");
                String name = rs.getString("student_name");
                double marks = rs.getDouble("marks");
                double total = marks;
                double percentage = marks;
                String result;
                if (percentage >= 40) {
                    result = "Pass";
                }
                else {
                    result = "Fail";
```

```
            }
            System.out.println("Student ID: " + id);
            System.out.println("Student Name: " + name);
            System.out.println("Total Marks: " + total);
            System.out.println("Percentage: " + percentage + "%");
            System.out.println("Result: " + result);
            System.out.println("----------------------------");
        }
        con.close();
    }
    catch (Exception e) {
        System.out.println(e);
    }
  }
}
```

Q-21. Write a Java JDBC program to validate a username and password using records stored in a database.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;
public class LoginValidation {
   public static void main(String[] args) {
      String url = "jdbc:mysql://localhost:3306/studentdb";
      String user = "root";
      String password = "";
      Scanner sc = new Scanner(System.in);
      try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            String sql = "SELECT * FROM users WHERE username = ? AND password
= ?";
            PreparedStatement ps = con.prepareStatement(sql);
            System.out.print("Enter Username: ");
            String username = sc.nextLine();
            System.out.print("Enter Password: ");
            String pass = sc.nextLine();
            ps.setString(1, username);
            ps.setString(2, pass);
            ResultSet rs = ps.executeQuery();
            if (rs.next()) {
                System.out.println("Login successful!");
            }
            else {
                System.out.println("Invalid username or password!");
            }
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Q-22. Write a Java JDBC program to search and display database records based on a name entered by the user.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
```

```
import java.sql.ResultSet;
import java.util.Scanner;
public class SearchByName {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        Scanner sc = new Scanner(System.in);
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            String sql = "SELECT * FROM student WHERE student_name LIKE ?";
            PreparedStatement ps = con.prepareStatement(sql);
            System.out.print("Enter Student Name: ");
            String name = sc.nextLine();
            ps.setString(1, "%" + name + "%");
            ResultSet rs = ps.executeQuery();
            boolean found = false;
            while (rs.next()) {
                found = true;
                System.out.println("Student ID: " + rs.getInt("student_id"));
                System.out.println("Name: " + rs.getString("student_name"));
                System.out.println("Course: " + rs.getString("course"));
                System.out.println("Semester: " + rs.getInt("semester"));
                System.out.println("Marks: " + rs.getDouble("marks"));
                System.out.println("----------------------------");
            }
            if (!found) {
                System.out.println("No student found!");
```

```
            }
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Q-23. Write a Java JDBC program to maintain bank account details such as Account Number, Customer Name, Account Type, and Balance and perform database operations.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.Statement;
import java.util.Scanner;
public class BankAccount {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        Scanner sc = new Scanner(System.in);
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();
```

```
        String create = "CREATE TABLE IF NOT EXISTS bank_account (account_no
INT PRIMARY KEY, customer_name VARCHAR(50), account_type VARCHAR(30),
balance DOUBLE)";
        stmt.executeUpdate(create);
        String insert = "INSERT INTO bank_account VALUES (?, ?, ?, ?)";
        PreparedStatement ps = con.prepareStatement(insert);
        System.out.print("Enter Account Number: ");
        int accountNo = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Customer Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Account Type: ");
        String type = sc.nextLine();
        System.out.print("Enter Balance: ");
        double balance = sc.nextDouble();
        ps.setInt(1, accountNo);
        ps.setString(2, name);
        ps.setString(3, type);
        ps.setDouble(4, balance);
        ps.executeUpdate();
        System.out.println("Account inserted successfully!");
        ResultSet rs = stmt.executeQuery("SELECT * FROM bank_account");
        System.out.println("\nAccount No\tCustomer Name\tAccount Type\tBalance");
        while (rs.next()) {
            System.out.println(rs.getInt("account_no") + "\t\t" +
rs.getString("customer_name") + "\t\t" + rs.getString("account_type") + "\t\t" +
rs.getDouble("balance"));
        }
        con.close();
    }
```

```
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Q-24. Write a Java JDBC program to maintain library book details such as Book ID, Book Name, Author, Publisher, and Price and perform database operations.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.Statement;
import java.util.Scanner;
public class LibraryBook {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "";
        Scanner sc = new Scanner(System.in);
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();
            String create = "CREATE TABLE IF NOT EXISTS book (book_id INT
PRIMARY KEY, book_name VARCHAR(100), author VARCHAR(50), publisher
VARCHAR(50), price DOUBLE)";
            stmt.executeUpdate(create);
            String insert = "INSERT INTO book VALUES (?, ?, ?, ?, ?)";
            PreparedStatement ps = con.prepareStatement(insert);
```

```
            System.out.print("Enter Book ID: ");
            int id = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter Book Name: ");
            String name = sc.nextLine();
            System.out.print("Enter Author: ");
            String author = sc.nextLine();
            System.out.print("Enter Publisher: ");
            String publisher = sc.nextLine();
            System.out.print("Enter Price: ");
            double price = sc.nextDouble();
            ps.setInt(1, id);
            ps.setString(2, name);
            ps.setString(3, author);
            ps.setString(4, publisher);
            ps.setDouble(5, price);
            ps.executeUpdate();
            System.out.println("Book inserted successfully!");
            ResultSet rs = stmt.executeQuery("SELECT * FROM book");
            System.out.println("\nID\tBook Name\tAuthor\tPublisher\tPrice");
            System.out.println("-------------------------------------------------------");
            while (rs.next()) {
                System.out.println(rs.getInt("book_id") + "\t" + rs.getString("book_name") +
"\t" + rs.getString("author") + "\t" + rs.getString("publisher") + "\t" +
rs.getDouble("price"));
            }
            con.close();
        }
        catch (Exception e) {
            System.out.println(e);
```

```
        }
    }
}
```

Q-25. Develop a Java JDBC-based database application for managing student records with proper database connectivity, data insertion, retrieval, searching, updating, and deletion facilities.

```
import java.sql.Connection;
                    sc.nextLine();
                    System.out.print("Enter Student Name: ");
                    String name = sc.nextLine();
                    System.out.print("Enter Course: ");
                    String course = sc.nextLine();
                    System.out.print("Enter Semester: ");
                    int semester = sc.nextInt();
                    System.out.print("Enter Marks: ");
                    double marks = sc.nextDouble();
                    ps1.setInt(1, id);
                    ps1.setString(2, name);
                    ps1.setString(3, course);
                    ps1.setInt(4, semester);
                    ps1.setDouble(5, marks);
                    ps1.executeUpdate();
                    System.out.println("Student inserted successfully!");
                    break;
                case 2:
                    PreparedStatement ps2 = con.prepareStatement("SELECT * FROM
student");
                    ResultSet rs = ps2.executeQuery();
                    System.out.println("ID\tName\tCourse\tSemester\tMarks");
                    while (rs.next()) {
```

```
                System.out.println(rs.getInt("student_id") + "\t" +
rs.getString("student_name") + "\t" + rs.getString("course") + "\t" +
rs.getInt("semester") + "\t\t" + rs.getDouble("marks"));
            }
            break;
        case 3:
            PreparedStatement ps3 = con.prepareStatement("SELECT * FROM
student WHERE student_id = ?");
            System.out.print("Enter Student ID: ");
            int searchId = sc.nextInt();
            ps3.setInt(1, searchId);
            ResultSet rs2 = ps3.executeQuery();
            if (rs2.next()) {
                System.out.println("Student ID: " + rs2.getInt("student_id"));
                System.out.println("Name: " + rs2.getString("student_name"));
                System.out.println("Course: " + rs2.getString("course"));
                System.out.println("Semester: " + rs2.getInt("semester"));
                System.out.println("Marks: " + rs2.getDouble("marks"));
            }
            else {
                System.out.println("Student not found!");
            }
            break;
        case 4:
            PreparedStatement ps4 = con.prepareStatement("UPDATE student
SET marks = ? WHERE student_id = ?");
            System.out.print("Enter Student ID: ");
            int updateId = sc.nextInt();
            System.out.print("Enter New Marks: ");
            double newMarks = sc.nextDouble();
```

```
                ps4.setDouble(1, newMarks);
                ps4.setInt(2, updateId);
                int updated = ps4.executeUpdate();
                if (updated > 0) {
                    System.out.println("Marks updated successfully!");
                }
                else {
                    System.out.println("Student not found!");
                }
                break;
            case 5:
                PreparedStatement ps5 = con.prepareStatement("DELETE FROM
student WHERE student_id = ?");
                System.out.print("Enter Student ID: ");
                int deleteId = sc.nextInt();
                ps5.setInt(1, deleteId);
                int deleted = ps5.executeUpdate();
                if (deleted > 0) {
                    System.out.println("Student deleted successfully!");
                }
                else {
                    System.out.println("Student not found!");
                }
                break;
            case 6:
                System.out.println("Program ended.");
                break;
            default:
                System.out.println("Invalid choice!");
```

```
            }
        } while (choice != 6);
        con.close();
    }
    catch (Exception e) {
        System.out.println(e);
    }
  }
}
```

Q-26. Write a Servlet program to accept student name and marks through an HTML form and display the total marks, percentage, grade, and result.

student.html

```
<html>
<body>
<form action="result" method="post">
Student Name: <input type="text" name="name"><br>
Subject 1: <input type="text" name="m1"><br>
Subject 2: <input type="text" name="m2"><br>
Subject 3: <input type="text" name="m3"><br>
<input type="submit" value="Calculate Result">
</form>
</body>
</html>
```

ResultServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
```

```
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/result")
public class ResultServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String name = request.getParameter("name");
        int m1 = Integer.parseInt(request.getParameter("m1"));
        int m2 = Integer.parseInt(request.getParameter("m2"));
        int m3 = Integer.parseInt(request.getParameter("m3"));
        int total = m1 + m2 + m3;
        double percentage = total / 3.0;
        String grade;
        String result;
        if (percentage >= 80) {
            grade = "A";
        }
        else if (percentage >= 60) {
            grade = "B";
        }
        else if (percentage >= 40) {
            grade = "C";
        }
        else {
            grade = "F";
        }
        if (percentage >= 40) {
```

```
            result = "Pass";
        }
        else {
            result = "Fail";
        }
        out.println("<h2>Student Result</h2>");
        out.println("Student Name: " + name + "<br>");
        out.println("Total Marks: " + total + "<br>");
        out.println("Percentage: " + percentage + "%<br>");
        out.println("Grade: " + grade + "<br>");
        out.println("Result: " + result);
    }
}
```

Q-27. Write a Servlet program to accept a number and determine whether it is even, odd, prime, or composite.

number.html

```
<html>
<body>
<form action="number" method="post">
Enter Number: <input type="text" name="num">
<input type="submit" value="Check">
</form>
</body>
</html>
```

NumberServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
```

```
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/number")
public class NumberServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        int num = Integer.parseInt(request.getParameter("num"));
        boolean prime = true;
        if (num <= 1) {
            prime = false;
        }
        else {
            for (int i = 2; i <= num / 2; i++) {
                if (num % i == 0) {
                    prime = false;
                    break;
                }
            }
        }
        out.println("<h2>Number Result</h2>");
        if (num % 2 == 0) {
            out.println("Number is Even<br>");
        }
        else {
            out.println("Number is Odd<br>");
        }
```

```
        if (num > 1 && prime) {
            out.println("Number is Prime");
        }
        else if (num > 1) {
            out.println("Number is Composite");
        }
        else {
            out.println("Number is neither Prime nor Composite");
        }
    }
}
```

Q-28. Write a Servlet program to accept a string and display its reverse, number of characters, number of vowels, and whether it is a palindrome.

string.html

```
<html>
<body>
<form action="string" method="post">
Enter String: <input type="text" name="text">
<input type="submit" value="Check">
</form>
</body>
</html>
```

StringServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
```

```
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/string")
public class StringServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String str = request.getParameter("text");
        String reverse = new StringBuilder(str).reverse().toString();
        int vowels = 0;
        for (int i = 0; i < str.length(); i++) {
            char ch = Character.toLowerCase(str.charAt(i));
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
                vowels++;
            }
        }
        out.println("<h2>String Result</h2>");
        out.println("Original String: " + str + "<br>");
        out.println("Reverse: " + reverse + "<br>");
        out.println("Number of Characters: " + str.length() + "<br>");
        out.println("Number of Vowels: " + vowels + "<br>");
        if (str.equalsIgnoreCase(reverse)) {
            out.println("Palindrome: Yes");
        }
        else {
            out.println("Palindrome: No");
        }
    }
}
```

Q-29. Write a Servlet program to demonstrate the complete Servlet life cycle using init(), service(), and destroy() methods.

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletConfig;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
public class LifeCycleServlet extends HttpServlet {
    public void init(ServletConfig config) throws ServletException {
        System.out.println("Servlet initialized");
    }
    protected void service(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<h2>Servlet Life Cycle</h2>");
        out.println("service() method executed");
        System.out.println("service() method executed");
    }
    public void destroy() {
        System.out.println("Servlet destroyed");
    }
}
```

Q-30. Write a Servlet program to demonstrate RequestDispatcher using both forward() and include() methods.

ForwardServlet.java

```
import java.io.IOException;
import jakarta.servlet.RequestDispatcher;
```

```
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/forward")
public class ForwardServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        RequestDispatcher rd = request.getRequestDispatcher("/target");
        rd.forward(request, response);
    }
}
```

IncludeServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.RequestDispatcher;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/include")
public class IncludeServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<h2>Before Include</h2>");
        RequestDispatcher rd = request.getRequestDispatcher("/target");
```

```
        rd.include(request, response);
        out.println("<h2>After Include</h2>");
    }
}
```

TargetServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/target")
public class TargetServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        PrintWriter out = response.getWriter();
        out.println("<h2>Target Servlet Executed</h2>");
    }
}
```

Q-31. Write a Servlet program to demonstrate HTTP request and response handling by displaying request parameters, request method, URL, protocol, client IP address, and HTTP header information.

request.html

```
<html>
<body>
<form action="request" method="post">
Name: <input type="text" name="name">
<input type="submit" value="Submit">
</form>
```

```
</body>
</html>
```

RequestServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.util.Enumeration;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/request")
public class RequestServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<h2>HTTP Request Information</h2>");
        out.println("Name: " + request.getParameter("name") + "<br>");
        out.println("Request Method: " + request.getMethod() + "<br>");
        out.println("URL: " + request.getRequestURL() + "<br>");
        out.println("Protocol: " + request.getProtocol() + "<br>");
        out.println("Client IP: " + request.getRemoteAddr() + "<br>");
        out.println("<h3>HTTP Headers</h3>");
        Enumeration<String> headers = request.getHeaderNames();
        while (headers.hasMoreElements()) {
            String header = headers.nextElement();
            out.println(header + ": " + request.getHeader(header) + "<br>");
        }
```

```
    }
}
```

Q-32. Develop a Servlet-based login application using HttpSession that authenticates a user, maintains the user's session across multiple pages, displays the logged-in username, and provides a logout facility.

login.html

```
<html>
<body>
<form action="login" method="post">
Username: <input type="text" name="username"><br>
Password: <input type="password" name="password"><br>
<input type="submit" value="Login">
</form>
</body>
</html>
```

LoginServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/login")
public class LoginServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
        String username = request.getParameter("username");
        String password = request.getParameter("password");
```

```
        if (username.equals("admin") && password.equals("1234")) {
            HttpSession session = request.getSession();
            session.setAttribute("username", username);
            response.sendRedirect("home");
        }
        else {
            response.setContentType("text/html");
            PrintWriter out = response.getWriter();
            out.println("Invalid username or password!");
        }
    }
}
```

HomeServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/home")
public class HomeServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        HttpSession session = request.getSession(false);
        if (session != null && session.getAttribute("username") != null) {
```

```
            String username = (String) session.getAttribute("username");
            out.println("<h2>Welcome " + username + "</h2>");
            out.println("<a href='logout'>Logout</a>");
        }
        else {
            out.println("Please login first!");
        }
    }
}
```

LogoutServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/logout")
public class LogoutServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        HttpSession session = request.getSession(false);
        if (session != null) {
            session.invalidate();
        }
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<h2>Logout successful!</h2>");
```

```
        out.println("<a href='login.html'>Login Again</a>");
    }
}
```

Q-33. Develop a Servlet-based application to demonstrate Cookie-based session tracking by creating, retrieving, updating, and deleting cookies.

CookieServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/cookie")
public class CookieServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        Cookie cookie = new Cookie("username", "Dhruvi");
        cookie.setMaxAge(60 * 60);
        response.addCookie(cookie);
        out.println("<h2>Cookie Created</h2>");
        out.println("Cookie Name: username<br>");
        out.println("Cookie Value: Dhruvi<br>");
        Cookie[] cookies = request.getCookies();
        if (cookies != null) {
            for (Cookie c : cookies) {
```

```
            out.println("Retrieved Cookie: " + c.getName() + " = " + c.getValue() +
"<br>");
          }
       }
       Cookie update = new Cookie("username", "Student");
       update.setMaxAge(60 * 60);
       response.addCookie(update);
       out.println("Cookie updated successfully!<br>");
       Cookie delete = new Cookie("username", "");
       delete.setMaxAge(0);
       response.addCookie(delete);
       out.println("Cookie deleted successfully!");
   }
}
```

Q-34. Develop a Servlet application that demonstrates multiple session-tracking techniques, including HttpSession, Cookies, and URL Rewriting, and displays the user information maintained by each technique.

SessionTrackingServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/tracking")
public class SessionTrackingServlet extends HttpServlet {
```

```
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String username = "Dhruvi";
        HttpSession session = request.getSession();
        session.setAttribute("username", username);
        Cookie cookie = new Cookie("username", username);
        cookie.setMaxAge(60 * 60);
        response.addCookie(cookie);
        String url = response.encodeURL("tracking?user=" + username);
        out.println("<h2>Session Tracking Techniques</h2>");
        out.println("<h3>HttpSession</h3>");
        out.println("Username: " + session.getAttribute("username") + "<br>");
        out.println("<h3>Cookies</h3>");
        out.println("Cookie created for username: " + username + "<br>");
        out.println("<h3>URL Rewriting</h3>");
        out.println("<a href='" + url + "'>URL Rewriting Link</a>");
    }
}
```

Q-35. Develop a Servlet-based student result management application that accepts marks through an HTML form, validates the input, calculates total, percentage, grade, and result, and displays the formatted result.

result.html

```
<html>
<body>
<h2>Student Result</h2>
<form action="studentResult" method="post">
Student Name: <input type="text" name="name"><br>
Subject 1 Marks: <input type="text" name="m1"><br>
```

```
Subject 2 Marks: <input type="text" name="m2"><br>
Subject 3 Marks: <input type="text" name="m3"><br>
<input type="submit" value="Calculate Result">
</form>
</body>
</html>
```

StudentResultServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/studentResult")
public class StudentResultServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String name = request.getParameter("name");
        String s1 = request.getParameter("m1");
        String s2 = request.getParameter("m2");
        String s3 = request.getParameter("m3");
        try {
            if (name == null || name.trim().isEmpty() || s1 == null || s2 == null || s3 == null)
{
                out.println("<h2>Invalid Input!</h2>");
                return;
            }
```

```
int m1 = Integer.parseInt(s1);
int m2 = Integer.parseInt(s2);
int m3 = Integer.parseInt(s3);
if (m1 < 0 || m1 > 100 || m2 < 0 || m2 > 100 || m3 < 0 || m3 > 100) {
    out.println("<h2>Marks must be between 0 and 100!</h2>");
    return;
}
int total = m1 + m2 + m3;
double percentage = total / 3.0;
String grade;
String result;
if (m1 < 40 || m2 < 40 || m3 < 40) {
    result = "Fail";
    grade = "F";
}
else if (percentage >= 80) {
    grade = "A";
    result = "Pass";
}
else if (percentage >= 60) {
    grade = "B";
    result = "Pass";
}
else {
    grade = "C";
    result = "Pass";
}
out.println("<h2>Student Result</h2>");
out.println("Student Name: " + name + "<br>");
```

```
            out.println("Subject 1: " + m1 + "<br>");
            out.println("Subject 2: " + m2 + "<br>");
            out.println("Subject 3: " + m3 + "<br>");
            out.println("Total Marks: " + total + "<br>");
            out.println("Percentage: " + String.format("%.2f", percentage) + "%<br>");
            out.println("Grade: " + grade + "<br>");
            out.println("Result: " + result);
        }
        catch (NumberFormatException e) {
            out.println("<h2>Invalid marks! Please enter numbers only.</h2>");
        }
    }
}
```

Q-36. Write a Servlet program to establish a JDBC database connection and perform insertion and retrieval of student records.

student.html

```
<html>
<body>
<h2>Student Form</h2>
<form action="student" method="post">
Student ID: <input type="text" name="id"><br>
Student Name: <input type="text" name="name"><br>
Course: <input type="text" name="course"><br>
Semester: <input type="text" name="semester"><br>
Marks: <input type="text" name="marks"><br>
<input type="submit" value="Insert">
</form>
<br>
```

```
<a href="student">Display Students</a>
</body>
</html>
```

StudentServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/student")
public class StudentServlet extends HttpServlet {
    String url = "jdbc:mysql://localhost:3306/studentdb";
    String user = "root";
    String password = "";
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            String sql = "INSERT INTO student VALUES (?, ?, ?, ?, ?)";
            PreparedStatement ps = con.prepareStatement(sql);
```

```
            ps.setInt(1, Integer.parseInt(request.getParameter("id")));
            ps.setString(2, request.getParameter("name"));
            ps.setString(3, request.getParameter("course"));
            ps.setInt(4, Integer.parseInt(request.getParameter("semester")));
            ps.setDouble(5, Double.parseDouble(request.getParameter("marks")));
            ps.executeUpdate();
            out.println("<h2>Student inserted successfully!</h2>");
            con.close();
        }
        catch (Exception e) {
            out.println(e);
        }
    }
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            PreparedStatement ps = con.prepareStatement("SELECT * FROM student");
            ResultSet rs = ps.executeQuery();
            out.println("<h2>Student Records</h2>");
            out.println("<table
border='1'><tr><th>ID</th><th>Name</th><th>Course</th><th>Semester</th><th>
Marks</th></tr>");
            while (rs.next()) {
                out.println("<tr><td>" + rs.getInt("student_id") + "</td><td>" +
rs.getString("student_name") + "</td><td>" + rs.getString("course") + "</td><td>" +
rs.getInt("semester") + "</td><td>" + rs.getDouble("marks") + "</td></tr>");
            }
```

```
            out.println("</table>");
            con.close();
        }
        catch (Exception e) {
            out.println(e);
        }
    }
}
```

Q-37. Develop a Servlet-based application to insert and display student records using HTML, Servlet, JDBC, and MySQL.

student.html

```
<html>
<body>
<h2>Student Registration</h2>
<form action="students" method="post">
Student ID: <input type="text" name="id"><br>
Student Name: <input type="text" name="name"><br>
Course: <input type="text" name="course"><br>
Semester: <input type="text" name="semester"><br>
Marks: <input type="text" name="marks"><br>
<input type="submit" value="Save Student">
</form>
<br>
<a href="students">View Students</a>
</body>
</html>
```

StudentServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
```

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/students")
public class StudentServlet extends HttpServlet {
    String url = "jdbc:mysql://localhost:3306/studentdb";
    String user = "root";
    String password = "";
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            PreparedStatement ps = con.prepareStatement("INSERT INTO student
VALUES (?, ?, ?, ?, ?)");
            ps.setInt(1, Integer.parseInt(request.getParameter("id")));
            ps.setString(2, request.getParameter("name"));
            ps.setString(3, request.getParameter("course"));
            ps.setInt(4, Integer.parseInt(request.getParameter("semester")));
            ps.setDouble(5, Double.parseDouble(request.getParameter("marks")));
            ps.executeUpdate();
            out.println("<h2>Student record inserted successfully!</h2>");
```

```
            out.println("<a href='students'>Display Records</a>");
            con.close();
        }
        catch (Exception e) {
            out.println(e);
        }
    }
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            PreparedStatement ps = con.prepareStatement("SELECT * FROM student");
            ResultSet rs = ps.executeQuery();
            out.println("<h2>Student Records</h2>");
            out.println("<table
border='1'><tr><th>ID</th><th>Name</th><th>Course</th><th>Semester</th><th>
Marks</th></tr>");
            while (rs.next()) {
                out.println("<tr><td>" + rs.getInt(1) + "</td><td>" + rs.getString(2) +
"</td><td>" + rs.getString(3) + "</td><td>" + rs.getInt(4) + "</td><td>" +
rs.getDouble(5) + "</td></tr>");
            }
            out.println("</table>");
            con.close();
        }
        catch (Exception e) {
            out.println(e);
        }
```

```
    }
}
```

Q-38. Develop a Servlet-based application to search, update, and delete student records using Student ID and JDBC.

student.html

```
<html>
<body>
<h2>Student Management</h2>
<form action="manageStudent" method="post">
Student ID: <input type="text" name="id"><br>
Student Name: <input type="text" name="name"><br>
Marks: <input type="text" name="marks"><br>
<input type="submit" name="action" value="Search">
<input type="submit" name="action" value="Update">
<input type="submit" name="action" value="Delete">
</form>
</body>
</html>
```

ManageStudentServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
```

```
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/manageStudent")
public class ManageStudentServlet extends HttpServlet {
    String url = "jdbc:mysql://localhost:3306/studentdb";
    String user = "root";
    String password = "";
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String action = request.getParameter("action");
        int id = Integer.parseInt(request.getParameter("id"));
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            if (action.equals("Search")) {
                PreparedStatement ps = con.prepareStatement("SELECT * FROM student
WHERE student_id=?");
                ps.setInt(1, id);
                ResultSet rs = ps.executeQuery();
                if (rs.next()) {
                    out.println("Student ID: " + rs.getInt("student_id") + "<br>");
                    out.println("Name: " + rs.getString("student_name") + "<br>");
                    out.println("Course: " + rs.getString("course") + "<br>");
                    out.println("Semester: " + rs.getInt("semester") + "<br>");
                    out.println("Marks: " + rs.getDouble("marks"));
                }
                else {
                    out.println("Student not found!");
                }
```

```
        }
        else if (action.equals("Update")) {
            PreparedStatement ps = con.prepareStatement("UPDATE student SET
student_name=?, marks=? WHERE student_id=?");
            ps.setString(1, request.getParameter("name"));
            ps.setDouble(2, Double.parseDouble(request.getParameter("marks")));
            ps.setInt(3, id);
            int result = ps.executeUpdate();
            out.println(result > 0 ? "Student updated successfully!" : "Student not
found!");
        }
        else if (action.equals("Delete")) {
            PreparedStatement ps = con.prepareStatement("DELETE FROM student
WHERE student_id=?");
            ps.setInt(1, id);
            int result = ps.executeUpdate();
            out.println(result > 0 ? "Student deleted successfully!" : "Student not
found!");
        }
        con.close();
    }
    catch (Exception e) {
        out.println(e);
    }
  }
}
```

Q-39. Develop a Servlet-based CRUD application for managing student records using HTML, Servlet, JDBC, PreparedStatement, and MySQL.

student.html

```
<html>
<body>
```

```
<h2>Student CRUD Application</h2>
<form action="crud" method="post">
Student ID: <input type="text" name="id"><br>
Student Name: <input type="text" name="name"><br>
Course: <input type="text" name="course"><br>
Semester: <input type="text" name="semester"><br>
Marks: <input type="text" name="marks"><br>
<input type="submit" name="action" value="Insert">
<input type="submit" name="action" value="Update">
<input type="submit" name="action" value="Delete">
<input type="submit" name="action" value="Display">
</form>
</body>
</html>
```

StudentCRUDServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/crud")
public class StudentCRUDServlet extends HttpServlet {
    String url = "jdbc:mysql://localhost:3306/studentdb";
```

```
    String user = "root";
    String password = "";
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String action = request.getParameter("action");
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            if (action.equals("Insert")) {
                PreparedStatement ps = con.prepareStatement("INSERT INTO student
VALUES (?, ?, ?, ?, ?)");
                ps.setInt(1, Integer.parseInt(request.getParameter("id")));
                ps.setString(2, request.getParameter("name"));
                ps.setString(3, request.getParameter("course"));
                ps.setInt(4, Integer.parseInt(request.getParameter("semester")));
                ps.setDouble(5, Double.parseDouble(request.getParameter("marks")));
                ps.executeUpdate();
                out.println("Student inserted successfully!");
            }
            else if (action.equals("Update")) {
                PreparedStatement ps = con.prepareStatement("UPDATE student SET
student_name=?, course=?, semester=?, marks=? WHERE student_id=?");
                ps.setString(1, request.getParameter("name"));
                ps.setString(2, request.getParameter("course"));
                ps.setInt(3, Integer.parseInt(request.getParameter("semester")));
                ps.setDouble(4, Double.parseDouble(request.getParameter("marks")));
                ps.setInt(5, Integer.parseInt(request.getParameter("id")));
```

```
            out.println(ps.executeUpdate() > 0 ? "Student updated successfully!" :
"Student not found!");
        }
        else if (action.equals("Delete")) {
            PreparedStatement ps = con.prepareStatement("DELETE FROM student
WHERE student_id=?");
            ps.setInt(1, Integer.parseInt(request.getParameter("id")));
            out.println(ps.executeUpdate() > 0 ? "Student deleted successfully!" :
"Student not found!");
        }
        else if (action.equals("Display")) {
            PreparedStatement ps = con.prepareStatement("SELECT * FROM
student");
            ResultSet rs = ps.executeQuery();
            out.println("<h2>Student Records</h2>");
            out.println("<table
border='1'><tr><th>ID</th><th>Name</th><th>Course</th><th>Semester</th><th>
Marks</th></tr>");
            while (rs.next()) {
                out.println("<tr><td>" + rs.getInt(1) + "</td><td>" + rs.getString(2) +
"</td><td>" + rs.getString(3) + "</td><td>" + rs.getInt(4) + "</td><td>" +
rs.getDouble(5) + "</td></tr>");
            }
            out.println("</table>");
        }
        con.close();
    }
    catch (Exception e) {
        out.println(e);
    }
  }
}
```

Q-40. Develop a Servlet-based Employee Management System that allows the user to add, search, update, delete, and display employee records using Servlet, JDBC, and MySQL.

employee.html

```
<html>
<body>
<h2>Employee Management</h2>
<form action="employee" method="post">
Employee ID: <input type="text" name="id"><br>
Name: <input type="text" name="name"><br>
Department: <input type="text" name="department"><br>
Designation: <input type="text" name="designation"><br>
Salary: <input type="text" name="salary"><br>
<input type="submit" name="action" value="Add">
<input type="submit" name="action" value="Search">
<input type="submit" name="action" value="Update">
<input type="submit" name="action" value="Delete">
<input type="submit" name="action" value="Display">
</form>
</body>
</html>
```

EmployeeServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
```

```
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/employee")
public class EmployeeServlet extends HttpServlet {
    String url = "jdbc:mysql://localhost:3306/studentdb";
    String user = "root";
    String password = "";
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String action = request.getParameter("action");
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            int id = Integer.parseInt(request.getParameter("id"));
            if (action.equals("Add")) {
                PreparedStatement ps = con.prepareStatement("INSERT INTO employee
VALUES (?, ?, ?, ?, ?)");
                ps.setInt(1, id);
                ps.setString(2, request.getParameter("name"));
                ps.setString(3, request.getParameter("department"));
                ps.setString(4, request.getParameter("designation"));
                ps.setDouble(5, Double.parseDouble(request.getParameter("salary")));
                ps.executeUpdate();
                out.println("Employee added successfully!");
            }
            else if (action.equals("Search")) {
```

```
            PreparedStatement ps = con.prepareStatement("SELECT * FROM
employee WHERE employee_id=?");

            ps.setInt(1, id);

            ResultSet rs = ps.executeQuery();

            if (rs.next()) {

                out.println("ID: " + rs.getInt(1) + "<br>Name: " + rs.getString(2) +
"<br>Department: " + rs.getString(3) + "<br>Designation: " + rs.getString(4) +
"<br>Salary: " + rs.getDouble(5));

            }

            else {

                out.println("Employee not found!");

            }

        }

        else if (action.equals("Update")) {

            PreparedStatement ps = con.prepareStatement("UPDATE employee SET
name=?, department=?, designation=?, salary=? WHERE employee_id=?");

            ps.setString(1, request.getParameter("name"));

            ps.setString(2, request.getParameter("department"));

            ps.setString(3, request.getParameter("designation"));

            ps.setDouble(4, Double.parseDouble(request.getParameter("salary")));

            ps.setInt(5, id);

            out.println(ps.executeUpdate() > 0 ? "Employee updated successfully!" :
"Employee not found!");

        }

        else if (action.equals("Delete")) {

            PreparedStatement ps = con.prepareStatement("DELETE FROM
employee WHERE employee_id=?");

            ps.setInt(1, id);

            out.println(ps.executeUpdate() > 0 ? "Employee deleted successfully!" :
"Employee not found!");

        }
```

```
        else if (action.equals("Display")) {
            PreparedStatement ps = con.prepareStatement("SELECT * FROM
employee");
            ResultSet rs = ps.executeQuery();
            out.println("<h2>Employee Records</h2>");
            while (rs.next()) {
                out.println(rs.getInt(1) + " | " + rs.getString(2) + " | " + rs.getString(3) + " |
" + rs.getString(4) + " | " + rs.getDouble(5) + "<br>");
            }
        }
        con.close();
    }
    catch (Exception e) {
        out.println(e);
    }
  }
}
```

Q-41. Develop a Servlet-based Bank Account Management System that maintains account details and provides facilities for creating accounts, searching accounts, updating account information, and displaying account details using JDBC.

bank.html

```
<html>
<body>
<h2>Bank Account Management</h2>
<form action="bank" method="post">
Account Number: <input type="text" name="account"><br>
Customer Name: <input type="text" name="name"><br>
Account Type: <input type="text" name="type"><br>
Balance: <input type="text" name="balance"><br>
<input type="submit" name="action" value="Create">
```

```
<input type="submit" name="action" value="Search">
<input type="submit" name="action" value="Update">
<input type="submit" name="action" value="Display">
</form>
</body>
</html>
```

BankServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/bank")
public class BankServlet extends HttpServlet {
    String url = "jdbc:mysql://localhost:3306/studentdb";
    String user = "root";
    String password = "";
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String action = request.getParameter("action");
        try {
```

```
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection con = DriverManager.getConnection(url, user, password);
        int account = Integer.parseInt(request.getParameter("account"));
        if (action.equals("Create")) {
            PreparedStatement ps = con.prepareStatement("INSERT INTO
bank_account VALUES (?, ?, ?, ?)");
            ps.setInt(1, account);
            ps.setString(2, request.getParameter("name"));
            ps.setString(3, request.getParameter("type"));
            ps.setDouble(4, Double.parseDouble(request.getParameter("balance")));
            ps.executeUpdate();
            out.println("Account created successfully!");
        }
        else if (action.equals("Search")) {
            PreparedStatement ps = con.prepareStatement("SELECT * FROM
bank_account WHERE account_no=?");
            ps.setInt(1, account);
            ResultSet rs = ps.executeQuery();
            if (rs.next()) {
                out.println("Account Number: " + rs.getInt(1) + "<br>");
                out.println("Customer Name: " + rs.getString(2) + "<br>");
                out.println("Account Type: " + rs.getString(3) + "<br>");
                out.println("Balance: " + rs.getDouble(4));
            }
            else {
                out.println("Account not found!");
            }
        }
        else if (action.equals("Update")) {
```

```
            PreparedStatement ps = con.prepareStatement("UPDATE bank_account
SET customer_name=?, account_type=?, balance=? WHERE account_no=?");
            ps.setString(1, request.getParameter("name"));
            ps.setString(2, request.getParameter("type"));
            ps.setDouble(3, Double.parseDouble(request.getParameter("balance")));
            ps.setInt(4, account);
            out.println(ps.executeUpdate() > 0 ? "Account updated successfully!" :
"Account not found!");
        }
        else if (action.equals("Display")) {
            PreparedStatement ps = con.prepareStatement("SELECT * FROM
bank_account");
            ResultSet rs = ps.executeQuery();
            out.println("<h2>Bank Accounts</h2>");
            while (rs.next()) {
                out.println(rs.getInt(1) + " | " + rs.getString(2) + " | " + rs.getString(3) + " |
" + rs.getDouble(4) + "<br>");
            }
        }
        con.close();
    }
    catch (Exception e) {
        out.println(e);
    }
  }
}
```

Q-42. Develop a Servlet-based application that performs login authentication using database records, maintains the authenticated user's session, restricts access to authorized pages, and provides logout functionality.

login.html

```
<html>
```

```
<body>
<h2>Login</h2>
<form action="dbLogin" method="post">
Username: <input type="text" name="username"><br>
Password: <input type="password" name="password"><br>
<input type="submit" value="Login">
</form>
</body>
</html>
```

LoginServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/dbLogin")
public class LoginServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb", "root", "");
            PreparedStatement ps = con.prepareStatement("SELECT * FROM users
WHERE username=? AND password=?");
            ps.setString(1, request.getParameter("username"));
            ps.setString(2, request.getParameter("password"));
            ResultSet rs = ps.executeQuery();
            if (rs.next()) {
                HttpSession session = request.getSession();
                session.setAttribute("username", request.getParameter("username"));
                response.sendRedirect("dashboard");
            }
            else {
                out.println("Invalid username or password!");
            }
            con.close();
        }
        catch (Exception e) {
            out.println(e);
        }
    }
}
```

DashboardServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
```

```
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/dashboard")
public class DashboardServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        HttpSession session = request.getSession(false);
        if (session == null || session.getAttribute("username") == null) {
            response.sendRedirect("login.html");
            return;
        }
        out.println("<h2>Welcome " + session.getAttribute("username") + "</h2>");
        out.println("<p>You are authorized to access this page.</p>");
        out.println("<a href='logout'>Logout</a>");
    }
}
```

LogoutServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/logout")
public class LogoutServlet extends HttpServlet {
```

```
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        HttpSession session = request.getSession(false);
        if (session != null) {
            session.invalidate();
        }
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<h2>Logout Successful!</h2>");
        out.println("<a href='login.html'>Login Again</a>");
    }
}
```

Q-43. Develop a Servlet-based application to implement a search facility with multiple criteria, allowing users to search student records by Student ID, name, course, or semester using JDBC and PreparedStatement.

search.html

```
<html>
<body>
<h2>Search Student</h2>
<form action="searchStudent" method="post">
Search By:
<select name="type">
<option value="id">Student ID</option>
<option value="name">Name</option>
<option value="course">Course</option>
<option value="semester">Semester</option>
</select><br>
Enter Value: <input type="text" name="value">
<input type="submit" value="Search">
</form>
```

```
</body>
</html>
```

SearchStudentServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/searchStudent")
public class SearchStudentServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String type = request.getParameter("type");
        String value = request.getParameter("value");
        String sql;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb", "root", "");
            if (type.equals("id")) {
                sql = "SELECT * FROM student WHERE student_id=?";
            }
```

```
            else if (type.equals("name")) {
                sql = "SELECT * FROM student WHERE student_name LIKE ?";
            }
            else if (type.equals("course")) {
                sql = "SELECT * FROM student WHERE course=?";
            }
            else {
                sql = "SELECT * FROM student WHERE semester=?";
            }
            PreparedStatement ps = con.prepareStatement(sql);
            if (type.equals("id") || type.equals("semester")) {
                ps.setInt(1, Integer.parseInt(value));
            }
            else if (type.equals("name")) {
                ps.setString(1, "%" + value + "%");
            }
            else {
                ps.setString(1, value);
            }
            ResultSet rs = ps.executeQuery();
            out.println("<h2>Search Results</h2>");
            while (rs.next()) {
                out.println(rs.getInt(1) + " | " + rs.getString(2) + " | " + rs.getString(3) + " | "
+ rs.getInt(4) + " | " + rs.getDouble(5) + "<br>");
            }
            con.close();
        }
        catch (Exception e) {
            out.println(e);
```

```
    }
  }
}
```

Q-44. Develop a Servlet-based application to display student records with sorting and filtering options, such as sorting by name or marks and filtering students based on course or semester.

filter.html

```
<html>
<body>
<h2>Student Sorting and Filtering</h2>
<form action="filter" method="get">
Course: <input type="text" name="course"><br>
Semester: <input type="text" name="semester"><br>
Sort By:
<select name="sort">
<option value="student_name">Name</option>
<option value="marks">Marks</option>
</select>
<input type="submit" value="Display">
</form>
</body>
</html>
```

FilterServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
```

```
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/filter")
public class FilterServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String course = request.getParameter("course");
        String semester = request.getParameter("semester");
        String sort = request.getParameter("sort");
        if (!sort.equals("student_name") && !sort.equals("marks")) {
            sort = "student_name";
        }
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb", "root", "");
            String sql = "SELECT * FROM student WHERE course LIKE ? AND semester
LIKE ? ORDER BY " + sort;
            PreparedStatement ps = con.prepareStatement(sql);
            ps.setString(1, course == null || course.isEmpty() ? "%" : course);
            ps.setString(2, semester == null || semester.isEmpty() ? "%" : semester);
            ResultSet rs = ps.executeQuery();
            out.println("<h2>Student Records</h2>");
            while (rs.next()) {
                out.println(rs.getInt(1) + " | " + rs.getString(2) + " | " + rs.getString(3) + " | "
+ rs.getInt(4) + " | " + rs.getDouble(5) + "<br>");
```

```
            }
            con.close();
        }
        catch (Exception e) {
            out.println(e);
        }
    }
}
```

Q-45. Develop a Servlet-based application to generate a student marks report by retrieving records from MySQL and displaying total marks, percentage, grade, pass/fail status, and class-wise information.

StudentReportServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/report")
public class StudentReportServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        try {
```

```
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb", "root", "");
        PreparedStatement ps = con.prepareStatement("SELECT * FROM student");
        ResultSet rs = ps.executeQuery();
        out.println("<h2>Student Marks Report</h2>");
        out.println("<table
border='1'><tr><th>ID</th><th>Name</th><th>Course</th><th>Total</th><th>Perce
ntage</th><th>Grade</th><th>Result</th><th>Class</th></tr>");
        while (rs.next()) {
            int id = rs.getInt("student_id");
            String name = rs.getString("student_name");
            String course = rs.getString("course");
            double marks = rs.getDouble("marks");
            double percentage = marks;
            String grade;
            String result;
            String className;
            if (percentage >= 80) {
                grade = "A";
                className = "Distinction";
            }
            else if (percentage >= 60) {
                grade = "B";
                className = "First Class";
            }
            else if (percentage >= 50) {
                grade = "C";
                className = "Second Class";
            }
```

```
            else if (percentage >= 40) {
                grade = "D";
                className = "Pass Class";
            }
            else {
                grade = "F";
                className = "Fail";
            }
            result = percentage >= 40 ? "Pass" : "Fail";
            out.println("<tr><td>" + id + "</td><td>" + name + "</td><td>" + course +
"</td><td>" + marks + "</td><td>" + percentage + "%</td><td>" + grade +
"</td><td>" + result + "</td><td>" + className + "</td></tr>");
        }
        out.println("</table>");
        con.close();
    }
    catch (Exception e) {
        out.println(e);
    }
  }
}
```

Q-46. Develop a Servlet-based application using Servlet, JDBC, HttpSession, Cookies, and RequestDispatcher to implement a complete student management system with login, authentication, CRUD operations, session management, and logout.

login.html

```
<html>
<body>
<h2>Student Management Login</h2>
<form action="loginSystem" method="post">
Username: <input type="text" name="username"><br>
```

```
Password: <input type="password" name="password"><br>
<input type="submit" value="Login">
</form>
</body>
</html>
```

LoginSystemServlet.java

```
import java.io.IOException;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/loginSystem")
public class LoginSystemServlet extends HttpServlet {
   protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
       try {
           Class.forName("com.mysql.cj.jdbc.Driver");
           Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb", "root", "");
           PreparedStatement ps = con.prepareStatement("SELECT * FROM users
WHERE username=? AND password=?");
           ps.setString(1, request.getParameter("username"));
           ps.setString(2, request.getParameter("password"));
```

```
            ResultSet rs = ps.executeQuery();
            if (rs.next()) {
                HttpSession session = request.getSession();
                session.setAttribute("username", request.getParameter("username"));
                Cookie cookie = new Cookie("username",
request.getParameter("username"));
                cookie.setMaxAge(60 * 60);
                response.addCookie(cookie);
                request.getRequestDispatcher("/dashboard").forward(request, response);
            }
            else {
                response.getWriter().println("Invalid login!");
            }
            con.close();
        }
        catch (Exception e) {
            response.getWriter().println(e);
        }
    }
}
```

DashboardServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
```

```
@WebServlet("/dashboard")
public class DashboardServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        HttpSession session = request.getSession(false);
        if (session == null || session.getAttribute("username") == null) {
            response.sendRedirect("login.html");
            return;
        }
        out.println("<h2>Welcome " + session.getAttribute("username") + "</h2>");
        out.println("<a href='studentCRUD'>Manage Students</a><br>");
        out.println("<a href='logoutSystem'>Logout</a>");
    }
}
```

StudentCRUDServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
```

```
@WebServlet("/studentCRUD")
public class StudentCRUDServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        HttpSession session = request.getSession(false);
        if (session == null || session.getAttribute("username") == null) {
            response.sendRedirect("login.html");
            return;
        }
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb", "root", "");
            PreparedStatement ps = con.prepareStatement("SELECT * FROM student");
            ResultSet rs = ps.executeQuery();
            out.println("<h2>Student Records</h2>");
            while (rs.next()) {
                out.println(rs.getInt(1) + " | " + rs.getString(2) + " | " + rs.getString(3) + " | "
+ rs.getInt(4) + " | " + rs.getDouble(5) + "<br>");
            }
            con.close();
        }
        catch (Exception e) {
            out.println(e);
        }
    }
}
```

LogoutSystemServlet.java

```
import java.io.IOException;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/logoutSystem")
public class LogoutSystemServlet extends HttpServlet {
   protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
       HttpSession session = request.getSession(false);
       if (session != null) {
           session.invalidate();
       }
       Cookie cookie = new Cookie("username", "");
       cookie.setMaxAge(0);
       response.addCookie(cookie);
       response.sendRedirect("login.html");
   }
}
```

Q-47. Develop a complete Web-Based Student Management System using HTML, Servlet, JDBC, MySQL, and session management. The application should provide user login, authentication, student registration, search, display, update, delete, result calculation, session-based access control, and logout facilities.

login.html

```
<html>
<body>
<h2>Student Management System</h2>
```

```
<form action="login" method="post">
Username: <input type="text" name="username"><br>
Password: <input type="password" name="password"><br>
<input type="submit" value="Login">
</form>
</body>
</html>
```

StudentManagementServlet.java

```
import java.io.IOException;
        }
        catch (Exception e) {
            out.println(e);
        }
    }
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        HttpSession session = request.getSession(false);
        if (session == null || session.getAttribute("username") == null) {
            response.sendRedirect("login.html");
            return;
        }
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            manageStudent(request, out, con, request.getParameter("action"));
            con.close();
        }
```

```
        catch (Exception e) {
            out.println(e);
        }
    }
    private void manageStudent(HttpServletRequest request, PrintWriter out,
Connection con, String action) throws Exception {
        if (action == null) {
            action = "display";
        }
        if (action.equals("insert")) {
            PreparedStatement ps = con.prepareStatement("INSERT INTO student
VALUES (?, ?, ?, ?, ?)");
            ps.setInt(1, Integer.parseInt(request.getParameter("id")));
            ps.setString(2, request.getParameter("name"));
            ps.setString(3, request.getParameter("course"));
            ps.setInt(4, Integer.parseInt(request.getParameter("semester")));
            ps.setDouble(5, Double.parseDouble(request.getParameter("marks")));
            ps.executeUpdate();
            out.println("Student registered successfully!<br>");
        }
        else if (action.equals("search")) {
            PreparedStatement ps = con.prepareStatement("SELECT * FROM student
WHERE student_id=?");
            ps.setInt(1, Integer.parseInt(request.getParameter("id")));
            ResultSet rs = ps.executeQuery();
            if (rs.next()) {
                out.println("ID: " + rs.getInt(1) + "<br>");
                out.println("Name: " + rs.getString(2) + "<br>");
                out.println("Course: " + rs.getString(3) + "<br>");
                out.println("Semester: " + rs.getInt(4) + "<br>");
```

```
            out.println("Marks: " + rs.getDouble(5) + "<br>");
        }
        else {
            out.println("Student not found!");
        }
    }
    else if (action.equals("update")) {
        PreparedStatement ps = con.prepareStatement("UPDATE student SET
student_name=?, course=?, semester=?, marks=? WHERE student_id=?");
        ps.setString(1, request.getParameter("name"));
        ps.setString(2, request.getParameter("course"));
        ps.setInt(3, Integer.parseInt(request.getParameter("semester")));
        ps.setDouble(4, Double.parseDouble(request.getParameter("marks")));
        ps.setInt(5, Integer.parseInt(request.getParameter("id")));
        out.println(ps.executeUpdate() > 0 ? "Student updated successfully!" :
"Student not found!");
    }
    else if (action.equals("delete")) {
        PreparedStatement ps = con.prepareStatement("DELETE FROM student
WHERE student_id=?");
        ps.setInt(1, Integer.parseInt(request.getParameter("id")));
        out.println(ps.executeUpdate() > 0 ? "Student deleted successfully!" :
"Student not found!");
    }
    else if (action.equals("result")) {
        PreparedStatement ps = con.prepareStatement("SELECT * FROM student
WHERE student_id=?");
        ps.setInt(1, Integer.parseInt(request.getParameter("id")));
        ResultSet rs = ps.executeQuery();
        if (rs.next()) {
            double marks = rs.getDouble("marks");
```

```
            String grade = marks >= 80 ? "A" : marks >= 60 ? "B" : marks >= 40 ? "C" :
"F";
            String result = marks >= 40 ? "Pass" : "Fail";
            out.println("<h2>Student Result</h2>");
            out.println("Name: " + rs.getString("student_name") + "<br>");
            out.println("Total Marks: " + marks + "<br>");
            out.println("Percentage: " + marks + "%<br>");
            out.println("Grade: " + grade + "<br>");
            out.println("Result: " + result + "<br>");
        }
    }
    else {
        PreparedStatement ps = con.prepareStatement("SELECT * FROM student");
        ResultSet rs = ps.executeQuery();
        out.println("<h2>Student Records</h2>");
        while (rs.next()) {
            out.println(rs.getInt(1) + " | " + rs.getString(2) + " | " + rs.getString(3) + " | "
+ rs.getInt(4) + " | " + rs.getDouble(5) + "<br>");
        }
    }
  }
}
```

student.html

```
<html>
<body>
<h2>Student Management</h2>
<form action="studentManagement" method="post">
Student ID: <input type="text" name="id"><br>
Student Name: <input type="text" name="name"><br>
Course: <input type="text" name="course"><br>
```

```
Semester: <input type="text" name="semester"><br>
Marks: <input type="text" name="marks"><br>
<input type="submit" name="action" value="insert">
<input type="submit" name="action" value="search">
<input type="submit" name="action" value="update">
<input type="submit" name="action" value="delete">
<input type="submit" name="action" value="result">
<input type="submit" name="action" value="display">
</form>
<br>
<a href="logout">Logout</a>
</body>
</html>
```

LogoutServlet.java

```
import java.io.IOException;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/logout")
public class LogoutServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        HttpSession session = request.getSession(false);
        if (session != null) {
            session.invalidate();
        }
```

```
        response.sendRedirect("login.html");
    }
}
```

Q-48. Develop a Servlet-based Online Shopping Cart Application that allows a user to log in, view products retrieved from a database, add products to a cart, update quantities, remove products, calculate the total amount, and complete the checkout process using HttpSession and JDBC.

login.html

```
<html>
<body>
<h2>Online Shopping Login</h2>
<form action="shopLogin" method="post">
Username: <input type="text" name="username"><br>
Password: <input type="password" name="password"><br>
<input type="submit" value="Login">
</form>
</body>
</html>
```

ShopLoginServlet.java

```
import java.io.IOException;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
```

```
import jakarta.servlet.http.HttpSession;

@WebServlet("/shopLogin")

public class ShopLoginServlet extends HttpServlet {

   protected void doPost(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {

       try {

          Class.forName("com.mysql.cj.jdbc.Driver");

          Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");

          PreparedStatement ps=con.prepareStatement("SELECT * FROM users
WHERE username=? AND password=?");

          ps.setString(1,request.getParameter("username"));

          ps.setString(2,request.getParameter("password"));

          ResultSet rs=ps.executeQuery();

          if(rs.next()) {

             HttpSession session=request.getSession();

             session.setAttribute("username",request.getParameter("username"));

             response.sendRedirect("products");

          } else {

             response.getWriter().println("Invalid username or password!");

          }

          con.close();

       } catch(Exception e) {

          response.getWriter().println(e);

       }

   }

}
```

ProductServlet.java

```
import java.io.IOException;

import java.io.PrintWriter;
```

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/products")
public class ProductServlet extends HttpServlet {
   protected void doGet(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
       response.setContentType("text/html");
       PrintWriter out=response.getWriter();
       HttpSession session=request.getSession(false);
       if(session==null||session.getAttribute("username")==null) {
          response.sendRedirect("login.html");
          return;
       }
       try {
          Class.forName("com.mysql.cj.jdbc.Driver");
          Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
          PreparedStatement ps=con.prepareStatement("SELECT * FROM product");
          ResultSet rs=ps.executeQuery();
          out.println("<h2>Products</h2>");
          while(rs.next()) {
```

```
            out.println(rs.getInt("product_id")+" - "+rs.getString("product_name")+" -
Rs."+rs.getDouble("price"));
            out.println("<a href='cart?id="+rs.getInt("product_id")+"'> Add to
Cart</a><br>");
        }
        out.println("<br><a href='cart'>View Cart</a>");
        con.close();
    } catch(Exception e) {
        out.println(e);
    }
  }
}
```

CartServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.HashMap;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/cart")
public class CartServlet extends HttpServlet {
  protected void doGet(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
```

```
        response.setContentType("text/html");
        PrintWriter out=response.getWriter();
        HttpSession session=request.getSession();
        HashMap<Integer,Integer>
cart=(HashMap<Integer,Integer>)session.getAttribute("cart");
        if(cart==null) {
            cart=new HashMap<>();
            session.setAttribute("cart",cart);
        }
        String id=request.getParameter("id");
        if(id!=null) {
            int productId=Integer.parseInt(id);
            cart.put(productId,cart.getOrDefault(productId,0)+1);
        }
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
            double total=0;
            out.println("<h2>Shopping Cart</h2>");
            for(Integer productId:cart.keySet()) {
                PreparedStatement ps=con.prepareStatement("SELECT * FROM product
WHERE product_id=?");
                ps.setInt(1,productId);
                ResultSet rs=ps.executeQuery();
                if(rs.next()) {
                    double price=rs.getDouble("price");
                    int quantity=cart.get(productId);
                    double amount=price*quantity;
                    total+=amount;
```

```
                out.println(rs.getString("product_name")+" | Price: "+price+" | Quantity: "+quantity+" | Amount: "+amount+"<br>");
                out.println("<a href='cart?id="+productId+"'>Add One</a><br>");
            }
        }
        out.println("<h3>Total Amount: Rs."+total+"</h3>");
        out.println("<a href='products'>Continue Shopping</a><br>");
        out.println("<a href='checkout'>Checkout</a>");
        con.close();
    } catch(Exception e) {
        out.println(e);
    }
  }
}
```

CheckoutServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/checkout")
public class CheckoutServlet extends HttpServlet {
  protected void doGet(HttpServletRequest request,HttpServletResponse response)throws ServletException,IOException {
    response.setContentType("text/html");
    PrintWriter out=response.getWriter();
    HttpSession session=request.getSession(false);
```

```
        if(session==null||session.getAttribute("username")==null) {
            response.sendRedirect("login.html");
            return;
        }
        session.removeAttribute("cart");
        out.println("<h2>Checkout Successful!</h2>");
        out.println("<p>Thank you for shopping with us.</p>");
        out.println("<a href='products'>Continue Shopping</a>");
    }
}
```

Q-49. Develop a Servlet-based Employee Payroll Management System that stores employee details in a database and calculates gross salary, deductions, and net salary based on employee information.

employee.html

```
<html>
<body>
<h2>Employee Payroll Management</h2>
<form action="payroll" method="post">
Employee ID: <input type="text" name="id"><br>
Name: <input type="text" name="name"><br>
Department: <input type="text" name="department"><br>
Basic Salary: <input type="text" name="basic"><br>
<input type="submit" value="Calculate Payroll">
</form>
</body>
</html>
```

PayrollServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
```

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/payroll")
public class PayrollServlet extends HttpServlet {
   protected void doPost(HttpServletRequest request,HttpServletResponse response)throws ServletException,IOException {
       response.setContentType("text/html");
       PrintWriter out=response.getWriter();
       try {
          int id=Integer.parseInt(request.getParameter("id"));
          String name=request.getParameter("name");
          String department=request.getParameter("department");
          double basic=Double.parseDouble(request.getParameter("basic"));
          double allowance=basic*0.20;
          double gross=basic+allowance;
          double deduction=gross*0.10;
          double net=gross-deduction;
          Class.forName("com.mysql.cj.jdbc.Driver");
          Connection con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
          PreparedStatement ps=con.prepareStatement("INSERT INTO payroll VALUES(?,?,?,?,?,?,?)");
          ps.setInt(1,id);
          ps.setString(2,name);
```

```
            ps.setString(3,department);
            ps.setDouble(4,basic);
            ps.setDouble(5,gross);
            ps.setDouble(6,deduction);
            ps.setDouble(7,net);
            ps.executeUpdate();
            out.println("<h2>Employee Payroll</h2>");
            out.println("Employee ID: "+id+"<br>");
            out.println("Name: "+name+"<br>");
            out.println("Department: "+department+"<br>");
            out.println("Basic Salary: "+basic+"<br>");
            out.println("Gross Salary: "+gross+"<br>");
            out.println("Deduction: "+deduction+"<br>");
            out.println("Net Salary: "+net);
            con.close();
        } catch(Exception e) {
            out.println(e);
        }
    }
}
```

Q-50. Develop a Servlet-based Hospital Patient Management System that allows authorized users to register patients, search patient records, update patient information, and display patient details using Servlet, JDBC, and MySQL.

patient.html

```
<html>
<body>
<h2>Hospital Patient Management</h2>
<form action="patient" method="post">
```

```
Patient ID: <input type="text" name="id"><br>
Patient Name: <input type="text" name="name"><br>
Age: <input type="text" name="age"><br>
Disease: <input type="text" name="disease"><br>
Doctor: <input type="text" name="doctor"><br>
<input type="submit" name="action" value="Register">
<input type="submit" name="action" value="Search">
<input type="submit" name="action" value="Update">
<input type="submit" name="action" value="Display">
</form>
</body>
</html>
```

PatientServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/patient")
public class PatientServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
        response.setContentType("text/html");
```

```
        PrintWriter out=response.getWriter();
        String action=request.getParameter("action");
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
            int id=Integer.parseInt(request.getParameter("id"));
            if(action.equals("Register")) {
                PreparedStatement ps=con.prepareStatement("INSERT INTO patient
VALUES(?,?,?,?,?)");
                ps.setInt(1,id);
                ps.setString(2,request.getParameter("name"));
                ps.setInt(3,Integer.parseInt(request.getParameter("age")));
                ps.setString(4,request.getParameter("disease"));
                ps.setString(5,request.getParameter("doctor"));
                ps.executeUpdate();
                out.println("Patient registered successfully!");
            } else if(action.equals("Search")) {
                PreparedStatement ps=con.prepareStatement("SELECT * FROM patient
WHERE patient_id=?");
                ps.setInt(1,id);
                ResultSet rs=ps.executeQuery();
                if(rs.next()) {
                    out.println("ID: "+rs.getInt(1)+"<br>");
                    out.println("Name: "+rs.getString(2)+"<br>");
                    out.println("Age: "+rs.getInt(3)+"<br>");
                    out.println("Disease: "+rs.getString(4)+"<br>");
                    out.println("Doctor: "+rs.getString(5));
                } else {
                    out.println("Patient not found!");
```

```
            }
        } else if(action.equals("Update")) {
            PreparedStatement ps=con.prepareStatement("UPDATE patient SET
patient_name=?,age=?,disease=?,doctor=? WHERE patient_id=?");
            ps.setString(1,request.getParameter("name"));
            ps.setInt(2,Integer.parseInt(request.getParameter("age")));
            ps.setString(3,request.getParameter("disease"));
            ps.setString(4,request.getParameter("doctor"));
            ps.setInt(5,id);
            out.println(ps.executeUpdate()>0?"Patient updated successfully!":"Patient
not found!");
        } else if(action.equals("Display")) {
            PreparedStatement ps=con.prepareStatement("SELECT * FROM
patient");
            ResultSet rs=ps.executeQuery();
            out.println("<h2>Patient Records</h2>");
            while(rs.next()) {
                out.println(rs.getInt(1)+" | "+rs.getString(2)+" | "+rs.getInt(3)+" |
"+rs.getString(4)+" | "+rs.getString(5)+"<br>");
            }
        }
        con.close();
    } catch(Exception e) {
        out.println(e);
    }
  }
}
```

Q-51. Develop a Servlet-based Course Registration System that allows students to log in, view available courses, register for courses, cancel

registration, and display their registered courses using Servlet, JDBC, and HttpSession.

login.html

```
<html>
<body>
<h2>Student Login</h2>
<form action="courseLogin" method="post">
Username: <input type="text" name="username"><br>
Password: <input type="password" name="password"><br>
<input type="submit" value="Login">
</form>
</body>
</html>
```

CourseLoginServlet.java

```
import java.io.IOException;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/courseLogin")
public class CourseLoginServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
        try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
            PreparedStatement ps=con.prepareStatement("SELECT * FROM users
WHERE username=? AND password=?");
            ps.setString(1,request.getParameter("username"));
            ps.setString(2,request.getParameter("password"));
            ResultSet rs=ps.executeQuery();
            if(rs.next()) {
                HttpSession session=request.getSession();
                session.setAttribute("username",request.getParameter("username"));
                response.sendRedirect("courses");
            } else {
                response.getWriter().println("Invalid login!");
            }
            con.close();
        } catch(Exception e) {
            response.getWriter().println(e);
        }
    }
}
```

CourseServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
```

```
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/courses")
public class CourseServlet extends HttpServlet {
   protected void doGet(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
       response.setContentType("text/html");
       PrintWriter out=response.getWriter();
       HttpSession session=request.getSession(false);
       if(session==null||session.getAttribute("username")==null) {
          response.sendRedirect("login.html");
          return;
       }
       try {
          Class.forName("com.mysql.cj.jdbc.Driver");
          Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
          PreparedStatement ps=con.prepareStatement("SELECT * FROM course");
          ResultSet rs=ps.executeQuery();
          out.println("<h2>Available Courses</h2>");
          while(rs.next()) {
             out.println(rs.getInt(1)+" - "+rs.getString(2)+" <a
href='registerCourse?id="+rs.getInt(1)+"'>Register</a><br>");
          }
          out.println("<br><a href='registeredCourses'>My Courses</a>");
          con.close();
       } catch(Exception e) {
          out.println(e);
```

```
        }
    }
}
```

RegisterCourseServlet.java

```
import java.io.IOException;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/registerCourse")
public class RegisterCourseServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
        HttpSession session=request.getSession(false);
        if(session==null||session.getAttribute("username")==null) {
            response.sendRedirect("login.html");
            return;
        }
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
            PreparedStatement ps=con.prepareStatement("INSERT INTO
registration(username,course_id) VALUES(?,?)");
            ps.setString(1,(String)session.getAttribute("username"));
```

```
            ps.setInt(2,Integer.parseInt(request.getParameter("id")));
            ps.executeUpdate();
            response.getWriter().println("Course registered successfully!<br><a
href='courses'>Back</a>");
            con.close();
        } catch(Exception e) {
            response.getWriter().println(e);
        }
    }
}
```

RegisteredCoursesServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/registeredCourses")
public class RegisteredCoursesServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
        response.setContentType("text/html");
        PrintWriter out=response.getWriter();
        HttpSession session=request.getSession(false);
```

```
        if(session==null||session.getAttribute("username")==null) {
            response.sendRedirect("login.html");
            return;
        }
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
            PreparedStatement ps=con.prepareStatement("SELECT
course.course_id,course.course_name FROM registration JOIN course ON
registration.course_id=course.course_id WHERE registration.username=?");
            ps.setString(1,(String)session.getAttribute("username"));
            ResultSet rs=ps.executeQuery();
            out.println("<h2>Registered Courses</h2>");
            while(rs.next()) {
                out.println(rs.getInt(1)+" - "+rs.getString(2)+"<br>");
            }
            con.close();
        } catch(Exception e) {
            out.println(e);
        }
    }
}
```

Q-52. Develop a Servlet-based Online Feedback System that allows authenticated users to submit feedback, stores the feedback in a database, and provides an administrator facility to view and analyze submitted feedback.

feedback.html

```
<html>
<body>
<h2>Online Feedback</h2>
```

```
<form action="feedback" method="post">
Name: <input type="text" name="name"><br>
Rating:
<select name="rating">
<option value="5">Excellent</option>
<option value="4">Very Good</option>
<option value="3">Good</option>
<option value="2">Average</option>
<option value="1">Poor</option>
</select><br>
Comments:<br>
<textarea name="comments"></textarea><br>
<input type="submit" value="Submit Feedback">
</form>
</body>
</html>
```

FeedbackServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
```

```
@WebServlet("/feedback")
public class FeedbackServlet extends HttpServlet {
   protected void doPost(HttpServletRequest request,HttpServletResponse response)throws ServletException,IOException {
      response.setContentType("text/html");
      PrintWriter out=response.getWriter();
      HttpSession session=request.getSession(false);
      if(session==null||session.getAttribute("username")==null) {
         response.sendRedirect("login.html");
         return;
      }
      try {
         Class.forName("com.mysql.cj.jdbc.Driver");
         Connection con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
         PreparedStatement ps=con.prepareStatement("INSERT INTO feedback(username,name,rating,comments) VALUES(?,?,?,?)");
         ps.setString(1,(String)session.getAttribute("username"));
         ps.setString(2,request.getParameter("name"));
         ps.setInt(3,Integer.parseInt(request.getParameter("rating")));
         ps.setString(4,request.getParameter("comments"));
         ps.executeUpdate();
         out.println("<h2>Feedback submitted successfully!</h2>");
         con.close();
      } catch(Exception e) {
         out.println(e);
      }
   }
}
```

AdminFeedbackServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/adminFeedback")
public class AdminFeedbackServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
        response.setContentType("text/html");
        PrintWriter out=response.getWriter();
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
            PreparedStatement ps=con.prepareStatement("SELECT * FROM feedback");
            ResultSet rs=ps.executeQuery();
            out.println("<h2>Feedback Report</h2>");
            while(rs.next()) {
                out.println("Name: "+rs.getString("name")+" | Rating: "+rs.getInt("rating")+"
| Comments: "+rs.getString("comments")+"<br>");
            }
            PreparedStatement avg=con.prepareStatement("SELECT
AVG(rating),COUNT(*) FROM feedback");
```

```
            ResultSet ar=avg.executeQuery();
            if(ar.next()) {
                out.println("<h3>Average Rating: "+ar.getDouble(1)+"</h3>");
                out.println("<h3>Total Feedback: "+ar.getInt(2)+"</h3>");
            }
            con.close();
        } catch(Exception e) {
            out.println(e);
        }
    }
}
```

Q-53. Develop a Servlet-based Student Attendance Management System that allows faculty to record attendance, update attendance records, search attendance by student or date, and generate an attendance report using Servlet and JDBC.

attendance.html

```
<html>
<body>
<h2>Student Attendance Management</h2>
<form action="attendance" method="post">
Student ID: <input type="text" name="id"><br>
Date: <input type="date" name="date"><br>
Status:
<select name="status">
<option value="Present">Present</option>
<option value="Absent">Absent</option>
</select><br>
<input type="submit" name="action" value="Record">
<input type="submit" name="action" value="Update">
```

```
<input type="submit" name="action" value="Search">
<input type="submit" name="action" value="Report">
</form>
</body>
</html>
```

AttendanceServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/attendance")
public class AttendanceServlet extends HttpServlet {
   protected void doPost(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
       response.setContentType("text/html");
       PrintWriter out=response.getWriter();
       String action=request.getParameter("action");
       try {
          Class.forName("com.mysql.cj.jdbc.Driver");
          Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
          int id=Integer.parseInt(request.getParameter("id"));
          if(action.equals("Record")) {
```

```
            PreparedStatement ps=con.prepareStatement("INSERT INTO
attendance(student_id,attendance_date,status) VALUES(?,?,?)");
            ps.setInt(1,id);
            ps.setString(2,request.getParameter("date"));
            ps.setString(3,request.getParameter("status"));
            ps.executeUpdate();
            out.println("Attendance recorded successfully!");
        } else if(action.equals("Update")) {
            PreparedStatement ps=con.prepareStatement("UPDATE attendance SET
status=? WHERE student_id=? AND attendance_date=?");
            ps.setString(1,request.getParameter("status"));
            ps.setInt(2,id);
            ps.setString(3,request.getParameter("date"));
            out.println(ps.executeUpdate()>0?"Attendance updated
successfully!":"Record not found!");
        } else if(action.equals("Search")) {
            PreparedStatement ps=con.prepareStatement("SELECT * FROM
attendance WHERE student_id=?");
            ps.setInt(1,id);
            ResultSet rs=ps.executeQuery();
            while(rs.next()) {
                out.println("Student ID: "+rs.getInt(1)+" | Date: "+rs.getDate(2)+" |
Status: "+rs.getString(3)+"<br>");
            }
        } else if(action.equals("Report")) {
            PreparedStatement ps=con.prepareStatement("SELECT
student_id,COUNT(*) AS total,SUM(status='Present') AS present FROM attendance
GROUP BY student_id");
            ResultSet rs=ps.executeQuery();
            out.println("<h2>Attendance Report</h2>");
            while(rs.next()) {
```

```
                int total=rs.getInt("total");
                int present=rs.getInt("present");
                double percentage=(present*100.0)/total;
                out.println("Student ID: "+rs.getInt("student_id")+" | Total: "+total+" |
Present: "+present+" | Percentage: "+percentage+"%<br>");
            }
        }
        con.close();
    } catch(Exception e) {
        out.println(e);
    }
  }
}
```

Q-54. Develop a Servlet-based Result Management System that allows an administrator to enter and update student marks, while students can log in and view their result, percentage, grade, and pass/fail status.

result.html

```
<html>
<body>
<h2>Result Management</h2>
<form action="result" method="post">
Student ID: <input type="text" name="id"><br>
Student Name: <input type="text" name="name"><br>
Marks: <input type="text" name="marks"><br>
<input type="submit" name="action" value="Save">
<input type="submit" name="action" value="Update">
<input type="submit" name="action" value="View">
</form>
</body>
```

```
</html>
```

ResultServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/result")
public class ResultServlet extends HttpServlet {
   protected void doPost(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
       response.setContentType("text/html");
       PrintWriter out=response.getWriter();
       String action=request.getParameter("action");
       try {
          Class.forName("com.mysql.cj.jdbc.Driver");
          Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
          int id=Integer.parseInt(request.getParameter("id"));
          if(action.equals("Save")) {
             PreparedStatement ps=con.prepareStatement("INSERT INTO result
VALUES(?,?,?)");
             ps.setInt(1,id);
             ps.setString(2,request.getParameter("name"));
```

```
            ps.setDouble(3,Double.parseDouble(request.getParameter("marks")));
            ps.executeUpdate();
            out.println("Result saved successfully!");
        } else if(action.equals("Update")) {
            PreparedStatement ps=con.prepareStatement("UPDATE result SET
marks=? WHERE student_id=?");
            ps.setDouble(1,Double.parseDouble(request.getParameter("marks")));
            ps.setInt(2,id);
            out.println(ps.executeUpdate()>0?"Marks updated successfully!":"Student
not found!");
        } else if(action.equals("View")) {
            PreparedStatement ps=con.prepareStatement("SELECT * FROM result
WHERE student_id=?");
            ps.setInt(1,id);
            ResultSet rs=ps.executeQuery();
            if(rs.next()) {
                double marks=rs.getDouble("marks");
                String grade;
                if(marks>=80) grade="A";
                else if(marks>=60) grade="B";
                else if(marks>=40) grade="C";
                else grade="F";
                String result=marks>=40?"Pass":"Fail";
                out.println("<h2>Student Result</h2>");
                out.println("Student ID: "+rs.getInt("student_id")+"<br>");
                out.println("Name: "+rs.getString("student_name")+"<br>");
                out.println("Total Marks: "+marks+"<br>");
                out.println("Percentage: "+marks+"%<br>");
                out.println("Grade: "+grade+"<br>");
                out.println("Result: "+result);
```

```
            } else {
                out.println("Result not found!");
            }
        }
        con.close();
    } catch(Exception e) {
        out.println(e);
    }
  }
}
```

Q-55. Develop a complete College Management System using HTML, Servlet, JDBC, MySQL, HttpSession, Cookies, and RequestDispatcher. The system should provide authentication and separate facilities for managing students, courses, attendance, examination results, and logout.

login.html

```
<html>
<body>
<h2>College Management System</h2>
<form action="collegeLogin" method="post">
Username: <input type="text" name="username"><br>
Password: <input type="password" name="password"><br>
<input type="submit" value="Login">
</form>
</body>
</html>
```

CollegeLoginServlet.java

```
import java.io.IOException;
import java.sql.Connection;
import java.sql.DriverManager;
```

```
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/collegeLogin")
public class CollegeLoginServlet extends HttpServlet {
   protected void doPost(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
       try {
          Class.forName("com.mysql.cj.jdbc.Driver");
          Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
          PreparedStatement ps=con.prepareStatement("SELECT * FROM users
WHERE username=? AND password=?");
          ps.setString(1,request.getParameter("username"));
          ps.setString(2,request.getParameter("password"));
          ResultSet rs=ps.executeQuery();
          if(rs.next()) {
             HttpSession session=request.getSession();
             session.setAttribute("username",request.getParameter("username"));
             Cookie cookie=new
Cookie("username",request.getParameter("username"));
             cookie.setMaxAge(3600);
             response.addCookie(cookie);
             request.getRequestDispatcher("/dashboard").forward(request,response);
```

```
            } else {
                response.getWriter().println("Invalid username or password!");
            }
            con.close();
        } catch(Exception e) {
            response.getWriter().println(e);
        }
    }
}
```

DashboardServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/dashboard")
public class DashboardServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
        response.setContentType("text/html");
        PrintWriter out=response.getWriter();
        HttpSession session=request.getSession(false);
        if(session==null||session.getAttribute("username")==null) {
            response.sendRedirect("login.html");
            return;
        }
```

```
        out.println("<h2>College Management System</h2>");
        out.println("Welcome: "+session.getAttribute("username")+"<br><br>");
        out.println("<a href='students'>Student Management</a><br>");
        out.println("<a href='courses'>Course Management</a><br>");
        out.println("<a href='attendance'>Attendance Management</a><br>");
        out.println("<a href='results'>Result Management</a><br>");
        out.println("<a href='collegeLogout'>Logout</a>");
    }
}
```

StudentServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/students")
public class StudentServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
        response.setContentType("text/html");
        PrintWriter out=response.getWriter();
        HttpSession session=request.getSession(false);
```

```
        if(session==null||session.getAttribute("username")==null) {
            response.sendRedirect("login.html");
            return;
        }
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
            PreparedStatement ps=con.prepareStatement("SELECT * FROM student");
            ResultSet rs=ps.executeQuery();
            out.println("<h2>Student Management</h2>");
            while(rs.next()) {
                out.println(rs.getInt(1)+" | "+rs.getString(2)+" | "+rs.getString(3)+" |
"+rs.getInt(4)+" | "+rs.getDouble(5)+"<br>");
            }
            con.close();
        } catch(Exception e) {
            out.println(e);
        }
    }
}
```

CourseServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
```

```
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/courses")
public class CourseServlet extends HttpServlet {
   protected void doGet(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
       response.setContentType("text/html");
       PrintWriter out=response.getWriter();
       HttpSession session=request.getSession(false);
       if(session==null||session.getAttribute("username")==null) {
          response.sendRedirect("login.html");
          return;
       }
       try {
          Class.forName("com.mysql.cj.jdbc.Driver");
          Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
          PreparedStatement ps=con.prepareStatement("SELECT * FROM course");
          ResultSet rs=ps.executeQuery();
          out.println("<h2>Course Management</h2>");
          while(rs.next()) {
             out.println(rs.getInt(1)+" | "+rs.getString(2)+"<br>");
          }
          con.close();
       } catch(Exception e) {
          out.println(e);
       }
   }
```

```
}
```

AttendanceServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/attendance")
public class AttendanceServlet extends HttpServlet {
   protected void doGet(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
       response.setContentType("text/html");
       PrintWriter out=response.getWriter();
       HttpSession session=request.getSession(false);
       if(session==null||session.getAttribute("username")==null) {
          response.sendRedirect("login.html");
          return;
       }
       try {
          Class.forName("com.mysql.cj.jdbc.Driver");
          Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");
```

```
        PreparedStatement ps=con.prepareStatement("SELECT
student_id,COUNT(*) AS total,SUM(status='Present') AS present FROM attendance
GROUP BY student_id");
        ResultSet rs=ps.executeQuery();
        out.println("<h2>Attendance Management</h2>");
        while(rs.next()) {
            int total=rs.getInt("total");
            int present=rs.getInt("present");
            double percentage=present*100.0/total;
            out.println("Student ID: "+rs.getInt("student_id")+" | Attendance:
"+percentage+"%<br>");
        }
        con.close();
    } catch(Exception e) {
        out.println(e);
    }
  }
}
```

ResultServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
```

```
import jakarta.servlet.http.HttpSession;

@WebServlet("/results")

public class ResultServlet extends HttpServlet {

   protected void doGet(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {

      response.setContentType("text/html");

      PrintWriter out=response.getWriter();

      HttpSession session=request.getSession(false);

      if(session==null||session.getAttribute("username")==null) {

         response.sendRedirect("login.html");

         return;

      }

      try {

         Class.forName("com.mysql.cj.jdbc.Driver");

         Connection
con=DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb","root","");

         PreparedStatement ps=con.prepareStatement("SELECT * FROM result");

         ResultSet rs=ps.executeQuery();

         out.println("<h2>Examination Results</h2>");

         while(rs.next()) {

            double marks=rs.getDouble("marks");

            String grade=marks>=80?"A":marks>=60?"B":marks>=40?"C":"F";

            String result=marks>=40?"Pass":"Fail";

            out.println("Student ID: "+rs.getInt("student_id")+" | Name:
"+rs.getString("student_name")+" | Marks: "+marks+" | Percentage: "+marks+"% |
Grade: "+grade+" | Result: "+result+"<br>");

         }

         con.close();

      } catch(Exception e) {

         out.println(e);
```

```
        }
    }
}
```

CollegeLogoutServlet.java

```
import java.io.IOException;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
@WebServlet("/collegeLogout")
public class CollegeLogoutServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request,HttpServletResponse
response)throws ServletException,IOException {
        HttpSession session=request.getSession(false);
        if(session!=null) {
            session.invalidate();
        }
        Cookie cookie=new Cookie("username","");
        cookie.setMaxAge(0);
        response.addCookie(cookie);
        response.sendRedirect("login.html");
    }
}
```

Enterprise Java Technology

Practical Assignment - II

Name: Darshan Prajapati
Roll No: 24
MCA_3

1. Create a JSP page to accept student details such as Student ID, Student Name, Course, Semester and Email, and display the submitted information on another JSP page.

student.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html>
<head>
   <title>Student Details Form</title>
</head>
<body>
   <h2>Student Details Form</h2>

   <form action="display.jsp" method="post">
      Student ID: <input type="text" name="studentId" required><br><br>

      Student Name: <input type="text" name="studentName" value="Dhruv" required><br><br>

      Course: <input type="text" name="course" value="MCA_01" required><br><br>

      Semester: <input type="text" name="semester" value="01" required><br><br>

      Email: <input type="email" name="email" required><br><br>

      <input type="submit" value="Submit">
      <input type="reset" value="Reset">
   </form>
</body>
</html>
```

display.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html>
<head>
   <title>Student Details</title>
</head>
<body>
   <h2>Submitted Student Details</h2>

   <%
      String studentId = request.getParameter("studentId");
      String studentName = request.getParameter("studentName");
      String course = request.getParameter("course");
      String semester = request.getParameter("semester");
      String email = request.getParameter("email");
   %>

   <table border="1" cellpadding="10">
      <tr>
         <th>Student ID</th>
         <td><%= studentId %></td>
      </tr>
      <tr>
         <th>Student Name</th>
         <td><%= studentName %></td>
      </tr>
      <tr>
         <th>Course</th>
         <td><%= course %></td>
      </tr>
      <tr>
         <th>Semester</th>
         <td><%= semester %></td>
      </tr>
      <tr>
         <th>Email</th>
         <td><%= email %></td>
      </tr>
   </table>

   <br>
   <a href="student.jsp">Back to Form</a>
</body>
</html>
```

2. Create a JSP page that accepts five subject marks and calculates the total, percentage and grade.

marks.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html>
<head>
   <title>Student Marks</title>
</head>
<body>
   <h2>Student Marks Calculator</h2>
   <h3>Name: Dhruv | Roll No: 01 | MCA_01</h3>

   <form action="result.jsp" method="post">
      Subject 1 Marks: <input type="number" name="m1" min="0" max="100" required><br><br>
      Subject 2 Marks: <input type="number" name="m2" min="0" max="100" required><br><br>
      Subject 3 Marks: <input type="number" name="m3" min="0" max="100" required><br><br>
      Subject 4 Marks: <input type="number" name="m4" min="0" max="100" required><br><br>
      Subject 5 Marks: <input type="number" name="m5" min="0" max="100" required><br><br>

      <input type="submit" value="Calculate Result">
      <input type="reset" value="Reset">
   </form>
</body>
</html>
```

result.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html>
<head>
   <title>Student Result</title>
</head>
<body>
   <h2>Student Result</h2>

   <%
      int m1 = Integer.parseInt(request.getParameter("m1"));
      int m2 = Integer.parseInt(request.getParameter("m2"));
```

```
        int m3 = Integer.parseInt(request.getParameter("m3"));
        int m4 = Integer.parseInt(request.getParameter("m4"));
        int m5 = Integer.parseInt(request.getParameter("m5"));

        int total = m1 + m2 + m3 + m4 + m5;
        double percentage = total / 5.0;
        String grade;

        if (m1 < 35 || m2 < 35 || m3 < 35 || m4 < 35 || m5 < 35) {
            grade = "Fail";
        } else if (percentage >= 90) {
            grade = "A+";
        } else if (percentage >= 80) {
            grade = "A";
        } else if (percentage >= 70) {
            grade = "B";
        } else if (percentage >= 60) {
            grade = "C";
        } else if (percentage >= 50) {
            grade = "D";
        } else {
            grade = "Pass";
        }
    %>

    <p><b>Name:</b> Dhruv</p>
    <p><b>Roll No:</b> 01</p>
    <p><b>Course:</b> MCA_01</p>

    <table border="1" cellpadding="10">
        <tr><th>Subject</th><th>Marks</th></tr>
        <tr><td>Subject 1</td><td><%= m1 %></td></tr>
        <tr><td>Subject 2</td><td><%= m2 %></td></tr>
        <tr><td>Subject 3</td><td><%= m3 %></td></tr>
        <tr><td>Subject 4</td><td><%= m4 %></td></tr>
        <tr><td>Subject 5</td><td><%= m5 %></td></tr>
        <tr><th>Total</th><th><%= total %> / 500</th></tr>
        <tr><th>Percentage</th><th><%= String.format("%.2f", percentage) %>%</th></tr>
        <tr><th>Grade</th><th><%= grade %></th></tr>
    </table>

    <br>
    <a href="marks.jsp">Back to Marks Form</a>
</body>
</html>
```

4. Create a JSP page that demonstrates the use of JSP declaration, scriptlet and expression tags.

index.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html>
<head>
   <title>JSP Tags Demonstration</title>
</head>
<body>
   <h2>JSP Declaration, Scriptlet and Expression Tags</h2>

   <%!
      String name = "Dhruv";
      int rollNo = 1;

      int square(int n) {
         return n * n;
      }
   %>

   <%
      String course = "MCA_01";
      int semester = 3;
      int number = 5;
   %>

   <h3>Student Details</h3>
   <p>Name: <%= name %></p>
   <p>Roll No: <%= rollNo %></p>
   <p>Course: <%= course %></p>
   <p>Semester: <%= semester %></p>

   <h3>Expression Tag Output</h3>
   <p>Number: <%= number %></p>
   <p>Square of <%= number %> is <%= square(number) %></p>

</body>
</html>
```

5. Create a JSP application to demonstrate the use of the following implicit objects:

request
response
session
application
Out

index.jsp

```
<%@ page language="java" %>
<html>
<head>
   <title>JSP Implicit Objects</title>
</head>
<body>
   <h2>JSP Implicit Objects</h2>

   <form method="post">
      Enter Name:
      <input type="text" name="name">
      <input type="submit" value="Submit">
   </form>

   <%
      String name = request.getParameter("name");

      if (name != null) {
         session.setAttribute("name", name);
         application.setAttribute("course", "MCA_01");

         out.println("<h3>Student Details</h3>");
         out.println("Request: " + name + "<br>");
         out.println("Response: " + response.getContentType() + "<br>");
         out.println("Session: " + session.getAttribute("name") + "<br>");
         out.println("Application: " + application.getAttribute("course") + "<br>");
         out.println("Out: Welcome " + name + "<br>");
         out.println("Roll No: 01");
      }
   %>

</body>
</html>
```

6. Create a JSP application that accepts username and password and displays a personalized welcome message using the request object.

```
<%@ page language="java" %>
<html>
<head>
   <title>Welcome Page</title>
</head>
<body>
   <h2>Login Form</h2>

   <form method="post">
      Username: <input type="text" name="username" required><br><br>
      Password: <input type="password" name="password" required><br><br>
      <input type="submit" value="Login">
   </form>

   <%
      String username = request.getParameter("username");
      String password = request.getParameter("password");

      if (username != null && password != null) {
   %>
      <h3>Welcome, <%= username %>!</h3>
      <p>You have successfully submitted your login details.</p>
   <%
      }
   %>

</body>
</html>
```

7. Create a JSP application that stores the logged-in user's username in a session and displays it on multiple JSP pages.

login.jsp

```
<%@ page language="java" %>
<html>
<body>
   <h2>Login Page</h2>
```

```
    <form action="welcome.jsp" method="post">
        Username: <input type="text" name="username" required><br><br>
        Password: <input type="password" name="password" required><br><br>
        <input type="submit" value="Login">
    </form>
</body>
</html>
```

welcome.jsp

```
<%@ page language="java" %>
<html>
<body>
    <%
        String username = request.getParameter("username");
        session.setAttribute("username", username);
    %>

    <h2>Welcome, <%= session.getAttribute("username") %>!</h2>

    <a href="profile.jsp">Go to Profile</a>
</body>
</html>
```

profile.jsp

```
<%@ page language="java" %>
<html>
<body>
    <h2>Profile Page</h2>

    <%
        String username = (String) session.getAttribute("username");

        if (username != null) {
    %>
        <h3>Logged-in User: <%= username %></h3>
    <%
        } else {
    %>
        <h3>Please login first!</h3>
        <a href="login.jsp">Login</a>
    <%
        }
    %>
```

```
    <br><br>
    <a href="welcome.jsp">Go to Welcome Page</a>
</body>
</html>
```

8. Create a JSP application using include to display a common header and footer on multiple JSP pages.

header.jsp

```
<h2>Welcome to MCA_01</h2>
<hr>
```

footer.jsp

```
<hr>
<p>Name: Dhruv | Roll No: 01</p>
<p>© 2026 MCA Student Portal</p>
```

index.jsp

```
<%@ page language="java" %>
<html>
<body>

    <%@ include file="header.jsp" %>

    <h3>Home Page</h3>
    <p>Welcome to the Home Page.</p>

    <%@ include file="footer.jsp" %>

</body>
</html>
```

about.jsp

```
<%@ page language="java" %>
<html>
<body>

    <%@ include file="header.jsp" %>
```

```
    <h3>About Page</h3>
    <p>This is the About Page.</p>

    <%@ include file="footer.jsp" %>

</body>
</html>
```

9. Create a JSP application using the JSP forward action to transfer a request from one JSP page to another.

index.jsp

```
<%@ page language="java" %>
<html>
<body>
    <h2>JSP Forward Example</h2>

    <form action="forward.jsp" method="post">
        Enter Name:
        <input type="text" name="name" required>
        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

forward.jsp

```
<%@ page language="java" %>
<%
    request.setAttribute("course", "MCA_01");
%>

<jsp:forward page="welcome.jsp" />
```

welcome.jsp

```
<%@ page language="java" %>
<html>
<body>
    <h2>Welcome Page</h2>
```

```
    <%
        String name = request.getParameter("name");
        String course = (String) request.getAttribute("course");
    %>

    <h3>Welcome, <%= name %>!</h3>
    <p>Course: <%= course %></p>
    <p>Name: Dhruv</p>
    <p>Roll No: 01</p>
</body>
</html>
```

10. Create a JSP page that uses JavaBean properties to accept and display student information.

Student.java

```
package mybean;

public class Student {
    private int id;
    private String name;
    private String course;
    private String email;

    public Student() {
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }
```

```
    public String getCourse() {
        return course;
    }

    public void setCourse(String course) {
        this.course = course;
    }

    public String getEmail() {
        return email;
    }

    public void setEmail(String email) {
        this.email = email;
    }
}
```

student.jsp

```
<%@ page language="java" %>
<html>
<body>
    <h2>Student Information</h2>

    <form action="display.jsp" method="post">
        Student ID: <input type="number" name="id" required><br><br>
        Student Name: <input type="text" name="name" value="Dhruv" required><br><br>
        Course: <input type="text" name="course" value="MCA_01" required><br><br>
        Email: <input type="email" name="email" required><br><br>

        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

display.jsp

```
<%@ page language="java" %>
<jsp:useBean id="s" class="mybean.Student" scope="request" />
<jsp:setProperty name="s" property="*" />

<html>
<body>
    <h2>Student Details</h2>

    Student ID: <jsp:getProperty name="s" property="id" /><br><br>
```

```
    Student Name: <jsp:getProperty name="s" property="name" /><br><br>
    Course: <jsp:getProperty name="s" property="course" /><br><br>
    Email: <jsp:getProperty name="s" property="email" /><br><br>

    Roll No: 01
</body>
</html>
```

11. Develop a JSP-based Student Result System that accepts marks of five subjects and displays:
Total Marks
Percentage
Grade
Pass/Fail Status
Use separate JSP pages for input and result.

input.jsp

```
<%@ page language="java" %>
<html>
<head>
    <title>Student Result System</title>
</head>
<body>
    <h2>Student Result System</h2>
    <h3>Name: Dhruv | Roll No: 01 | MCA_01</h3>

    <form action="result.jsp" method="post">
        Subject 1: <input type="number" name="m1" min="0" max="100" required><br><br>
        Subject 2: <input type="number" name="m2" min="0" max="100" required><br><br>
        Subject 3: <input type="number" name="m3" min="0" max="100" required><br><br>
        Subject 4: <input type="number" name="m4" min="0" max="100" required><br><br>
        Subject 5: <input type="number" name="m5" min="0" max="100" required><br><br>

        <input type="submit" value="Calculate Result">
        <input type="reset" value="Reset">
    </form>
</body>
</html>
```

result.jsp

```
<%@ page language="java" %>
<html>
<head>
   <title>Student Result</title>
</head>
<body>
   <h2>Student Result</h2>

   <%
      int m1 = Integer.parseInt(request.getParameter("m1"));
      int m2 = Integer.parseInt(request.getParameter("m2"));
      int m3 = Integer.parseInt(request.getParameter("m3"));
      int m4 = Integer.parseInt(request.getParameter("m4"));
      int m5 = Integer.parseInt(request.getParameter("m5"));

      int total = m1 + m2 + m3 + m4 + m5;
      double percentage = total / 5.0;

      String grade;
      String status;

      if (m1 < 35 || m2 < 35 || m3 < 35 || m4 < 35 || m5 < 35) {
         grade = "F";
         status = "Fail";
      } else {
         status = "Pass";

         if (percentage >= 90)
            grade = "A+";
         else if (percentage >= 80)
            grade = "A";
         else if (percentage >= 70)
            grade = "B";
         else if (percentage >= 60)
            grade = "C";
         else if (percentage >= 50)
            grade = "D";
         else
            grade = "E";
      }
   %>

   <p>Name: Dhruv</p>
   <p>Roll No: 01</p>
   <p>Course: MCA_01</p>

   <table border="1" cellpadding="10">
      <tr>
```

```
            <th>Subject</th>
            <th>Marks</th>
        </tr>
        <tr><td>Subject 1</td><td><%= m1 %></td></tr>
        <tr><td>Subject 2</td><td><%= m2 %></td></tr>
        <tr><td>Subject 3</td><td><%= m3 %></td></tr>
        <tr><td>Subject 4</td><td><%= m4 %></td></tr>
        <tr><td>Subject 5</td><td><%= m5 %></td></tr>
        <tr><th>Total Marks</th><td><%= total %> / 500</td></tr>
        <tr><th>Percentage</th><td><%= String.format("%.2f", percentage) %>%</td></tr>
        <tr><th>Grade</th><td><%= grade %></td></tr>
        <tr><th>Status</th><td><%= status %></td></tr>
    </table>

    <br>
    <a href="input.jsp">Back to Input Page</a>
</body>
</html>
```

12. Develop a JSP-based Login System using session management. After successful login, display a dashboard containing the username and provide a logout facility.

login.jsp

```
<%@ page language="java" %>
<html>
<head>
    <title>Login Page</title>
</head>
<body>
    <h2>Student Login</h2>

    <form action="validate.jsp" method="post">
        Username: <input type="text" name="username" required><br><br>
        Password: <input type="password" name="password" required><br><br>
        <input type="submit" value="Login">
    </form>
</body>
</html>
```

validate.jsp

```
<%@ page language="java" %>
```

```
<%
   String username = request.getParameter("username");
   String password = request.getParameter("password");

   if ("Dhruv".equals(username) && "1234".equals(password)) {
      session.setAttribute("username", username);
      response.sendRedirect("dashboard.jsp");
   } else {
      response.sendRedirect("login.jsp");
   }
%>
```

dashboard.jsp

```
<%@ page language="java" %>
<%
   String username = (String) session.getAttribute("username");

   if (username == null) {
      response.sendRedirect("login.jsp");
      return;
   }
%>
<html>
<head>
   <title>Dashboard</title>
</head>
<body>
   <h2>Student Dashboard</h2>

   <h3>Welcome, <%= username %>!</h3>
   <p>Name: Dhruv</p>
   <p>Roll No: 01</p>
   <p>Course: MCA_01</p>

   <a href="logout.jsp">Logout</a>
</body>
</html>
```

logout.jsp

```
<%@ page language="java" %>
<%
   session.invalidate();
   response.sendRedirect("login.jsp");
%>
```

13. Create a JSP application to demonstrate session tracking using:
Cookies
HttpSession
URL Rewriting

index.jsp

```
<%@ page language="java" %>
<html>
<body>
   <h2>Session Tracking</h2>

   <form action="tracking.jsp" method="post">
      Enter Name:
      <input type="text" name="username" required>
      <input type="submit" value="Submit">
   </form>
</body>
</html>
```

tracking.jsp

```
<%@ page language="java" %>
<%
   String username = request.getParameter("username");

   if (username != null) {
      Cookie c = new Cookie("username", username);
      c.setMaxAge(60 * 60);
      response.addCookie(c);

      session.setAttribute("username", username);
   } else {
      username = (String) session.getAttribute("username");
   }

   if (username == null) {
      response.sendRedirect("index.jsp");
      return;
   }

   String url = response.encodeURL("welcome.jsp");
```

```
%>

<html>
<body>
   <h2>Session Tracking Methods</h2>

   <p>Cookie: <%= username %></p>
   <p>HttpSession: <%= session.getAttribute("username") %></p>

   <a href="<%= url %>">Continue using URL Rewriting</a>
</body>
</html>
```

welcome.jsp

```
<%@ page language="java" %>
<%
   String username = (String) session.getAttribute("username");

   if (username == null) {
      response.sendRedirect("index.jsp");
      return;
   }
%>

<html>
<body>
   <h2>Welcome Page</h2>

   <p>Welcome, <%= username %>!</p>
   <p>URL Rewriting is demonstrated through the encoded URL.</p>

   <a href="index.jsp">Back to Home</a>
</body>
</html>
```

14. Develop a JSP application that retrieves student records from a database using JDBC and displays them in an HTML table.

student.jsp

```
<%@ page import="java.sql.*" %>
<%@ page language="java" %>
```

```
<html>
<head>
  <title>Student Records</title>
</head>
<body>
  <h2>Student Records</h2>
  <h3>Name: Dhruv | Roll No: 01 | MCA_01</h3>

  <table border="1" cellpadding="10">
    <tr>
      <th>ID</th>
      <th>Name</th>
      <th>Course</th>
      <th>Semester</th>
      <th>Email</th>
    </tr>

    <%
      try {
        Class.forName("com.mysql.cj.jdbc.Driver");

        Connection con = DriverManager.getConnection(
          "jdbc:mysql://localhost:3306/javadb",
          "root",
          "1234"
        );

        Statement st = con.createStatement();
        ResultSet rs = st.executeQuery("SELECT * FROM student");

        while (rs.next()) {
    %>
    <tr>
      <td><%= rs.getInt("id") %></td>
      <td><%= rs.getString("name") %></td>
      <td><%= rs.getString("course") %></td>
      <td><%= rs.getInt("semester") %></td>
      <td><%= rs.getString("email") %></td>
    </tr>
    <%
        }
        con.close();
      } catch (Exception e) {
        out.println("<tr><td colspan='5'>" + e.getMessage() + "</td></tr>");
      }
    %>
  </table>
```

```
</body>
</html>
```

15. Create a JSP application to perform CRUD operations on an Employee table using JSP and JDBC.

employee,.jsp

```
<%@ page import="java.sql.*" %>
<%@ page language="java" %>
<html>
<head>
   <title>Employee CRUD</title>
</head>
<body>
   <h2>Employee Management System</h2>
   <h3>Name: Dhruv | Roll No: 01 | MCA_01</h3>

   <form action="employee.jsp" method="post">
      Employee ID: <input type="number" name="id" required><br><br>
      Employee Name: <input type="text" name="name"><br><br>
      Department: <input type="text" name="department"><br><br>
      Salary: <input type="number" step="0.01" name="salary"><br><br>

      <button type="submit" name="action" value="insert">Insert</button>
      <button type="submit" name="action" value="update">Update</button>
      <button type="submit" name="action" value="delete">Delete</button>
   </form>

   <%
      Connection con = null;

      try {
         Class.forName("com.mysql.cj.jdbc.Driver");

         con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/javadb",
            "root",
            "1234"
         );

         String action = request.getParameter("action");

         if (action != null) {
            int id = Integer.parseInt(request.getParameter("id"));
```

```
        if (action.equals("insert")) {
            PreparedStatement ps = con.prepareStatement(
                "INSERT INTO employee VALUES (?, ?, ?, ?)"
            );
            ps.setInt(1, id);
            ps.setString(2, request.getParameter("name"));
            ps.setString(3, request.getParameter("department"));
            ps.setDouble(4, Double.parseDouble(request.getParameter("salary")));
            ps.executeUpdate();
            ps.close();
            out.println("<p>Employee inserted successfully!</p>");
        }

        else if (action.equals("update")) {
            PreparedStatement ps = con.prepareStatement(
                "UPDATE employee SET name=?, department=?, salary=? WHERE id=?"
            );
            ps.setString(1, request.getParameter("name"));
            ps.setString(2, request.getParameter("department"));
            ps.setDouble(3, Double.parseDouble(request.getParameter("salary")));
            ps.setInt(4, id);

            int rows = ps.executeUpdate();
            ps.close();
            out.println(rows > 0 ? "<p>Employee updated successfully!</p>"
                                 : "<p>Employee not found!</p>");
        }

        else if (action.equals("delete")) {
            PreparedStatement ps = con.prepareStatement(
                "DELETE FROM employee WHERE id=?"
            );
            ps.setInt(1, id);

            int rows = ps.executeUpdate();
            ps.close();
            out.println(rows > 0 ? "<p>Employee deleted successfully!</p>"
                                 : "<p>Employee not found!</p>");
        }
    }
%>

<h3>Employee Records</h3>

<table border="1" cellpadding="10">
  <tr>
    <th>ID</th>
```

```
            <th>Name</th>
            <th>Department</th>
            <th>Salary</th>
        </tr>

        <%
            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("SELECT * FROM employee");

            while (rs.next()) {
        %>
        <tr>
            <td><%= rs.getInt("id") %></td>
            <td><%= rs.getString("name") %></td>
            <td><%= rs.getString("department") %></td>
            <td><%= rs.getDouble("salary") %></td>
        </tr>
        <%
            }
            rs.close();
            st.close();
        } catch (Exception e) {
            out.println("<p>Error: " + e.getMessage() + "</p>");
        } finally {
            if (con != null) {
                try { con.close(); } catch (SQLException e) { }
            }
        }
        %>
    </table>
</body>
</html>
```

16. Create a JSP application that displays employee details based on a department selected from a dropdown list.

employee.jsp

```
<%@ page import="java.sql.*" %>
<%@ page language="java" %>

<html>
<head>
    <title>Employee Details</title>
</head>
```

```
<body>
    <h2>Employee Details by Department</h2>
    <h3>Name: Dhruv | Roll No: 01 | MCA_01</h3>

    <form method="post">
        Select Department:
        <select name="department">
            <option value="IT">IT</option>
            <option value="HR">HR</option>
            <option value="Finance">Finance</option>
        </select>
        <input type="submit" value="Search">
    </form>

    <br>

    <%
        String department = request.getParameter("department");

        if (department != null) {
            try {
                Class.forName("com.mysql.cj.jdbc.Driver");

                Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/javadb",
                    "root",
                    "1234"
                );

                PreparedStatement ps = con.prepareStatement(
                    "SELECT * FROM employee WHERE department = ?"
                );
                ps.setString(1, department);

                ResultSet rs = ps.executeQuery();
    %>

    <h3>Department: <%= department %></h3>

    <table border="1" cellpadding="10">
        <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Department</th>
            <th>Salary</th>
        </tr>

        <%
```

```
            boolean found = false;

            while (rs.next()) {
                found = true;
        %>
        <tr>
            <td><%= rs.getInt("id") %></td>
            <td><%= rs.getString("name") %></td>
            <td><%= rs.getString("department") %></td>
            <td><%= rs.getDouble("salary") %></td>
        </tr>
        <%
            }

            if (!found) {
        %>
        <tr>
            <td colspan="4">No employees found.</td>
        </tr>
        <%
            }

            rs.close();
            ps.close();
            con.close();

            } catch (Exception e) {
                out.println("Error: " + e.getMessage());
            }
        }
    %>
    </table>

</body>
</html>
```

17. Create a JSP page that uses Expression Language (EL) to display request parameters without using JSP scriptlets.

index.jsp

```
<%@ page language="java" %>
<html>
<head>
```

```
    <title>Expression Language Example</title>
</head>
<body>
    <h2>Student Details Form</h2>

    <form action="display.jsp" method="post">
        Student Name: <input type="text" name="name" value="Dhruv" required><br><br>
        Roll No: <input type="text" name="rollno" value="01" required><br><br>
        Course: <input type="text" name="course" value="MCA_01" required><br><br>

        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

display.jsp

```
<%@ page language="java" %>
<html>
<head>
    <title>Student Details</title>
</head>
<body>
    <h2>Student Details Using EL</h2>

    <p>Student Name: ${param.name}</p>
    <p>Roll No: ${param.rollno}</p>
    <p>Course: ${param.course}</p>

</body>
</html>
```

18. Create a JSP application using EL to display values stored in: Request scope Session scope Application scope

index.jsp

```
<%@ page language="java" %>
<%
    request.setAttribute("rname", "Dhruv");
    session.setAttribute("srollno", "01");
    application.setAttribute("course", "MCA_01");
```

```
%>

<html>
<body>
   <h2>EL Scope Example</h2>

   Request Scope: ${requestScope.rname}<br><br>
   Session Scope: ${sessionScope.srollno}<br><br>
   Application Scope: ${applicationScope.course}<br><br>
</body>
</html>
```

19. Create a JSP page that demonstrates EL operators for:
Arithmetic
Relational
Logical
Conditional

operator.jsp

```
<%@ page language="java" %>
<html>
<body>
   <h2>EL Operators Demonstration</h2>

   <h3>Arithmetic Operators</h3>
   Addition: ${10 + 5}<br>
   Subtraction: ${10 - 5}<br>
   Multiplication: ${10 * 5}<br>
   Division: ${10 / 5}<br><br>

   <h3>Relational Operators</h3>
   Equal: ${10 == 10}<br>
   Greater Than: ${10 > 5}<br>
   Less Than: ${5 < 10}<br><br>

   <h3>Logical Operators</h3>
   AND: ${true && true}<br>
   OR: ${true || false}<br>
   NOT: ${not false}<br><br>

   <h3>Conditional Operator</h3>
   Result: ${10 >= 5 ? "Pass" : "Fail"}<br><br>
```

```
</body>
</html>
```

20. Create a JSP application that uses EL to access properties of a JavaBean containing:
Student ID
Student Name
Course
Semester

Student.java

```
package mybean;

public class Student {
   private int id;
   private String name;
   private String course;
   private int semester;

   public Student() {
   }

   public int getId() {
      return id;
   }

   public void setId(int id) {
      this.id = id;
   }

   public String getName() {
      return name;
   }

   public void setName(String name) {
      this.name = name;
   }

   public String getCourse() {
      return course;
   }

   public void setCourse(String course) {
```

```
        this.course = course;
    }

    public int getSemester() {
        return semester;
    }

    public void setSemester(int semester) {
        this.semester = semester;
    }
}
```

student.jsp

```
<%@ page language="java" %>
<html>
<body>
    <h2>Student Information</h2>

    <form action="display.jsp" method="post">
        Student ID: <input type="number" name="id" required><br><br>
        Student Name: <input type="text" name="name" value="Dhruv" required><br><br>
        Course: <input type="text" name="course" value="MCA_01" required><br><br>
        Semester: <input type="number" name="semester" required><br><br>

        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

display.jsp

```
<%@ page language="java" %>
<jsp:useBean id="s" class="mybean.Student" scope="request" />
<jsp:setProperty name="s" property="*" />

<html>
<body>
    <h2>Student Details Using EL</h2>

    Student ID: ${s.id}<br><br>
    Student Name: ${s.name}<br><br>
    Course: ${s.course}<br><br>
    Semester: ${s.semester}<br><br>
</body>
</html>
```

21. Create a JSP page using EL to display values from a Java List and Map.

index.jsp

```
<%@ page import="java.util.*" %>
<%@ page language="java" %>

<%
   List<String> subjects = new ArrayList<>();
   subjects.add("Java");
   subjects.add("Python");
   subjects.add("JSP");

   Map<String, String> student = new HashMap<>();
   student.put("name", "Dhruv");
   student.put("rollno", "01");
   student.put("course", "MCA_01");

   request.setAttribute("subjects", subjects);
   request.setAttribute("student", student);
%>

<html>
<body>
   <h2>EL List and Map Example</h2>

   <h3>Student Details</h3>
   Name: ${student.name}<br>
   Roll No: ${student.rollno}<br>
   Course: ${student.course}<br>

   <h3>Subjects</h3>
   ${subjects[0]}<br>
   ${subjects[1]}<br>
   ${subjects[2]}<br>
</body>
</html>
```

22. Create a JSP application that uses EL to check whether a user is logged in and conditionally displays the user's dashboard or login message.

login.jsp

```
<%@ page language="java" %>
<html>
<body>
   <h2>Login</h2>

   <form action="dashboard.jsp" method="post">
      Username: <input type="text" name="username" required><br><br>
      <input type="submit" value="Login">
   </form>
</body>
</html>
```

dashboard.jsp

```
<%@ page language="java" %>

<%
   String username = request.getParameter("username");

   if (username != null && !username.trim().isEmpty()) {
      session.setAttribute("username", username);
   }
%>

<html>
<body>
   <h2>Student Dashboard</h2>

   <c:if test="${not empty sessionScope.username}">
      Welcome, ${sessionScope.username}!<br>
      Roll No: 01<br>
      Course: MCA_01
   </c:if>

   <c:if test="${empty sessionScope.username}">
      Please login first.
      <a href="login.jsp">Login</a>
   </c:if>
</body>
</html>
```

23. Develop a JSP application that accepts two numbers and performs arithmetic operations using EL instead of scriptlets.

index.jsp

```
<%@ page language="java" %>
<html>
<body>
   <h2>Arithmetic Calculator</h2>

   <form action="result.jsp" method="post">
      First Number: <input type="number" name="a" required><br><br>
      Second Number: <input type="number" name="b" required><br><br>
      <input type="submit" value="Calculate">
   </form>
</body>
</html>
```

result.jsp

```
<%@ page language="java" %>
<html>
<body>
   <h2>Arithmetic Results</h2>

   <p>Addition: ${param.a + param.b}</p>
   <p>Subtraction: ${param.a - param.b}</p>
   <p>Multiplication: ${param.a * param.b}</p>
   <p>Division: ${param.a / param.b}</p>
   <p>Remainder: ${param.a % param.b}</p>
</body>
</html>
```

24. Create a JSP application using JSTL <c:if> to determine whether a student has passed or failed based on percentage.

```
<%@ page language="java" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<html>
<body>
   <h2>Student Result</h2>

   <form method="post">
      Student Name: <input type="text" name="name" value="Dhruv" required><br><br>
      Roll No: <input type="text" name="rollno" value="01" required><br><br>
      Percentage: <input type="number" name="percentage"
                  min="0" max="100" step="0.01" required><br><br>
      <input type="submit" value="Check Result">
   </form>

   <c:if test="${not empty param.percentage}">
      <h3>Name: <c:out value="${param.name}"/></h3>
      <p>Roll No: <c:out value="${param.rollno}"/></p>
      <p>Percentage: <c:out value="${param.percentage}"/>%</p>

      <c:if test="${param.percentage >= 35}">
         <h3>Result: Pass</h3>
      </c:if>

      <c:if test="${param.percentage < 35}">
         <h3>Result: Fail</h3>
```

```
    </c:if>
  </c:if>
</body>
</html>
```

25. Create a JSP application using JSTL <c:choose>, <c:when> and <c:otherwise> to display grades based on student marks.

grade.jsp

```
<%@ page language="java" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<html>
<body>
  <h2>Student Grade System</h2>

  <form method="post">
    Student Name: <input type="text" name="name" value="Dhruv" required><br><br>
    Marks: <input type="number" name="marks" min="0" max="100" required><br><br>
    <input type="submit" value="Check Grade">
  </form>

  <c:if test="${not empty param.marks}">
    <p>Name: <c:out value="${param.name}"/></p>
    <p>Marks: <c:out value="${param.marks}"/></p>

    <c:choose>
      <c:when test="${param.marks >= 90}">Grade: A+</c:when>
      <c:when test="${param.marks >= 80}">Grade: A</c:when>
      <c:when test="${param.marks >= 70}">Grade: B</c:when>
      <c:when test="${param.marks >= 60}">Grade: C</c:when>
      <c:when test="${param.marks >= 35}">Grade: D</c:when>
      <c:otherwise>Grade: F (Fail)</c:otherwise>
    </c:choose>
  </c:if>
</body>
</html>
```

26. Create a JSP application using JSTL <c:forEach> to display a list of students in an HTML table.

students.jsp

```
<%@ page language="java" %>
<%@ page import="java.util.*" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<%
   List<String[]> students = new ArrayList<>();

   students.add(new String[]{"1", "Dhruv", "MCA_01"});
   students.add(new String[]{"2", "Raj", "MCA"});
   students.add(new String[]{"3", "Nirav", "MCA"});

   request.setAttribute("students", students);
%>

<html>
<body>
   <h2>Student Records</h2>

   <table border="1" cellpadding="10">
      <tr>
         <th>ID</th>
         <th>Name</th>
         <th>Course</th>
      </tr>

      <c:forEach var="s" items="${students}">
         <tr>
            <td><c:out value="${s[0]}"/></td>
            <td><c:out value="${s[1]}"/></td>
            <td><c:out value="${s[2]}"/></td>
         </tr>
      </c:forEach>
   </table>
</body>
</html>
```

27. Create a JSP application using JSTL <c:forEach> to display employee records retrieved from a database.

employee.jsp

```
<%@ page language="java" %>
<%@ page import="java.sql.*" %>
<%@ page import="java.util.*" %>
```

```
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<%
   List<Map<String, Object>> employees = new ArrayList<>();

   try {
      Class.forName("com.mysql.cj.jdbc.Driver");

      Connection con = DriverManager.getConnection(
         "jdbc:mysql://localhost:3306/javadb",
         "root",
         "1234"
      );

      Statement st = con.createStatement();
      ResultSet rs = st.executeQuery("SELECT * FROM employee");

      while (rs.next()) {
         Map<String, Object> emp = new HashMap<>();

         emp.put("id", rs.getInt("id"));
         emp.put("name", rs.getString("name"));
         emp.put("department", rs.getString("department"));
         emp.put("salary", rs.getDouble("salary"));

         employees.add(emp);
      }

      rs.close();
      st.close();
      con.close();

   } catch (Exception e) {
      request.setAttribute("error", e.getMessage());
   }

   request.setAttribute("employees", employees);
%>

<html>
<body>
   <h2>Employee Records</h2>

   <c:if test="${not empty error}">
      <p>Error: <c:out value="${error}"/></p>
   </c:if>

   <table border="1" cellpadding="10">
```

```
      <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Department</th>
        <th>Salary</th>
      </tr>

      <c:forEach var="e" items="${employees}">
        <tr>
          <td><c:out value="${e.id}"/></td>
          <td><c:out value="${e.name}"/></td>
          <td><c:out value="${e.department}"/></td>
          <td><c:out value="${e.salary}"/></td>
        </tr>
      </c:forEach>
    </table>
</body>
</html>
```

28. Create a JSP application demonstrating JSTL <c:set>, <c:remove> and <c:out>.

jstl.jsp

```
<%@ page language="java" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<html>
<body>
    <h2>JSTL Core Tags</h2>

    <h3>c:set Example</h3>
    <c:set var="name" value="Dhruv"/>
    <c:set var="course" value="MCA_01"/>

    Name: <c:out value="${name}"/><br><br>
    Course: <c:out value="${course}"/><br><br>

    <h3>c:remove Example</h3>
    <c:remove var="name"/>

    Name after removal:
    <c:out value="${name}" default="Name Removed"/><br><br>

    <h3>c:out Example</h3>
```

```
   Roll No: <c:out value="01"/><br><br>
</body>
</html>
```

29. Create a JSP application using JSTL and EL to display product information stored in a collection.

Display:
Product ID
Product Name
Quantity
Price
Total Price

products.jsp

```
<%@ page language="java" %>
<%@ page import="java.util.*" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<%
   List<Map<String, Object>> products = new ArrayList<>();

   Map<String, Object> p1 = new HashMap<>();
   p1.put("id", 101);
   p1.put("name", "Keyboard");
   p1.put("quantity", 2);
   p1.put("price", 500.0);
   products.add(p1);

   Map<String, Object> p2 = new HashMap<>();
   p2.put("id", 102);
   p2.put("name", "Mouse");
   p2.put("quantity", 3);
   p2.put("price", 250.0);
   products.add(p2);

   Map<String, Object> p3 = new HashMap<>();
   p3.put("id", 103);
   p3.put("name", "Monitor");
   p3.put("quantity", 1);
   p3.put("price", 5000.0);
   products.add(p3);
```

```
    request.setAttribute("products", products);
%>

<html>
<head>
    <title>Product Information</title>
</head>
<body>
    <h2>Product Information</h2>
    <h3>Name: Dhruv | Roll No: 01 | MCA_01</h3>

    <table border="1" cellpadding="10">
        <tr>
            <th>Product ID</th>
            <th>Product Name</th>
            <th>Quantity</th>
            <th>Price</th>
            <th>Total Price</th>
        </tr>

        <c:forEach var="p" items="${products}">
            <tr>
                <td><c:out value="${p.id}"/></td>
                <td><c:out value="${p.name}"/></td>
                <td><c:out value="${p.quantity}"/></td>
                <td><c:out value="${p.price}"/></td>
                <td>${p.quantity * p.price}</td>
            </tr>
        </c:forEach>
    </table>
</body>
</html>
```

30. Develop a Student Result Management System using JSP, JSTL and EL. The application should:
Accept student details and marks
Calculate total and percentage
Determine grade
Store student information in appropriate scope
Display the result using JSTL and EL
Avoid JSP scriptlets in the result page

input.jsp

```
<%@ page language="java" %>
<html>
<head>
   <title>Student Result System</title>
</head>
<body>
   <h2>Student Result Management System</h2>

   <form action="calculate.jsp" method="post">
      Student ID: <input type="text" name="id" required><br><br>
      Student Name: <input type="text" name="name" value="Dhruv" required><br><br>
      Roll No: <input type="text" name="rollno" value="01" required><br><br>
      Course: <input type="text" name="course" value="MCA_01" required><br><br>

      Subject 1: <input type="number" name="m1" min="0" max="100" required><br><br>
      Subject 2: <input type="number" name="m2" min="0" max="100" required><br><br>
      Subject 3: <input type="number" name="m3" min="0" max="100" required><br><br>
      Subject 4: <input type="number" name="m4" min="0" max="100" required><br><br>
      Subject 5: <input type="number" name="m5" min="0" max="100" required><br><br>

      <input type="submit" value="Calculate Result">
   </form>
</body>
</html>
```

calculate.jsp

```
<%@ page language="java" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:set var="m1" value="${param.m1}" />
<c:set var="m2" value="${param.m2}" />
<c:set var="m3" value="${param.m3}" />
<c:set var="m4" value="${param.m4}" />
<c:set var="m5" value="${param.m5}" />

<c:set var="total" value="${m1 + m2 + m3 + m4 + m5}" />
<c:set var="percentage" value="${total / 5}" />

<c:set var="grade" value="F" />
<c:set var="status" value="Fail" />

<c:if test="${m1 >= 35 and m2 >= 35 and m3 >= 35 and m4 >= 35 and m5 >= 35}">
   <c:set var="status" value="Pass" />
```

```
    <c:choose>
        <c:when test="${percentage >= 90}">
            <c:set var="grade" value="A+" />
        </c:when>
        <c:when test="${percentage >= 80}">
            <c:set var="grade" value="A" />
        </c:when>
        <c:when test="${percentage >= 70}">
            <c:set var="grade" value="B" />
        </c:when>
        <c:when test="${percentage >= 60}">
            <c:set var="grade" value="C" />
        </c:when>
        <c:when test="${percentage >= 50}">
            <c:set var="grade" value="D" />
        </c:when>
        <c:otherwise>
            <c:set var="grade" value="E" />
        </c:otherwise>
    </c:choose>
</c:if>

<c:set var="studentId" value="${param.id}" scope="session" />
<c:set var="studentName" value="${param.name}" scope="session" />
<c:set var="rollno" value="${param.rollno}" scope="session" />
<c:set var="course" value="${param.course}" scope="session" />
<c:set var="total" value="${total}" scope="session" />
<c:set var="percentage" value="${percentage}" scope="session" />
<c:set var="grade" value="${grade}" scope="session" />
<c:set var="status" value="${status}" scope="session" />

<c:redirect url="result.jsp" />
```

result.jsp

```
<%@ page language="java" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<html>
<head>
    <title>Student Result</title>
</head>
<body>
    <h2>Student Result</h2>

    <table border="1" cellpadding="10">
        <tr>
```

```
        <th>Student ID</th>
        <td><c:out value="${sessionScope.studentId}" /></td>
      </tr>
      <tr>
        <th>Student Name</th>
        <td><c:out value="${sessionScope.studentName}" /></td>
      </tr>
      <tr>
        <th>Roll No</th>
        <td><c:out value="${sessionScope.rollno}" /></td>
      </tr>
      <tr>
        <th>Course</th>
        <td><c:out value="${sessionScope.course}" /></td>
      </tr>
      <tr>
        <th>Total Marks</th>
        <td><c:out value="${sessionScope.total}" /> / 500</td>
      </tr>
      <tr>
        <th>Percentage</th>
        <td><c:out value="${sessionScope.percentage}" />%</td>
      </tr>
      <tr>
        <th>Grade</th>
        <td><c:out value="${sessionScope.grade}" /></td>
      </tr>
      <tr>
        <th>Status</th>
        <td><c:out value="${sessionScope.status}" /></td>
      </tr>
    </table>

    <br>
    <a href="input.jsp">Back to Input Page</a>
</body>
</html>
```

31. Develop an Employee Management System using JSP, JSTL, EL and JDBC with the following operations:

Add Employee

Display Employee

Search Employee

Update Employee
Delete Employee

employee.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<!DOCTYPE html>
<html>
<head>
  <title>Employee Management System</title>
</head>
<body>
  <h2>Employee Management System</h2>
  <p>Name: Dhruv | Roll No: 01 | MCA_01</p>

  <h3>Add Employee</h3>
  <form action="EmployeeServlet" method="post">
    <input type="hidden" name="action" value="add">

    Employee ID: <input type="number" name="id" required><br><br>
    Name: <input type="text" name="name" required><br><br>
    Department: <input type="text" name="department" required><br><br>
    Salary: <input type="number" name="salary" step="0.01" min="0" required><br><br>

    <input type="submit" value="Add Employee">
  </form>

  <hr>

  <h3>Search Employee</h3>
  <form action="EmployeeServlet" method="get">
    <input type="hidden" name="action" value="search">
    Employee ID: <input type="number" name="id" required>
    <input type="submit" value="Search">
  </form>

  <hr>

  <h3>Update Employee</h3>
  <form action="EmployeeServlet" method="post">
    <input type="hidden" name="action" value="update">

    Employee ID: <input type="number" name="id" required><br><br>
    Name: <input type="text" name="name" required><br><br>
```

```
    Department: <input type="text" name="department" required><br><br>
    Salary: <input type="number" name="salary" step="0.01" min="0" required><br><br>

    <input type="submit" value="Update Employee">
</form>

<hr>

<h3>Delete Employee</h3>
<form action="EmployeeServlet" method="post">
    <input type="hidden" name="action" value="delete">
    Employee ID: <input type="number" name="id" required>
    <input type="submit" value="Delete Employee">
</form>

<hr>

<h3>Display All Employees</h3>
<form action="EmployeeServlet" method="get">
    <input type="hidden" name="action" value="display">
    <input type="submit" value="Display Employees">
</form>

<c:if test="${not empty message}">
    <h3><c:out value="${message}"/></h3>
</c:if>

<c:if test="${not empty employee}">
    <h3>Search Result</h3>
    <table border="1" cellpadding="8">
        <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Department</th>
            <th>Salary</th>
        </tr>
        <tr>
            <td><c:out value="${employee.id}"/></td>
            <td><c:out value="${employee.name}"/></td>
            <td><c:out value="${employee.department}"/></td>
            <td><c:out value="${employee.salary}"/></td>
        </tr>
    </table>
</c:if>

<c:if test="${not empty employees}">
    <h3>Employee List</h3>
    <table border="1" cellpadding="8">
```

```
        <tr>
          <th>ID</th>
          <th>Name</th>
          <th>Department</th>
          <th>Salary</th>
        </tr>

        <c:forEach var="e" items="${employees}">
          <tr>
            <td><c:out value="${e.id}"/></td>
            <td><c:out value="${e.name}"/></td>
            <td><c:out value="${e.department}"/></td>
            <td><c:out value="${e.salary}"/></td>
          </tr>
        </c:forEach>
      </table>
    </c:if>
</body>
</html>
```

Employee.java

```
package com.example;

public class Employee {
    private int id;
    private String name;
    private String department;
    private double salary;

    public Employee() {
    }

    public Employee(int id, String name, String department, double salary) {
        this.id = id;
        this.name = name;
        this.department = department;
        this.salary = salary;
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
```

```
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getDepartment() {
        return department;
    }

    public void setDepartment(String department) {
        this.department = department;
    }

    public double getSalary() {
        return salary;
    }

    public void setSalary(double salary) {
        this.salary = salary;
    }
}
```

EmployeeServlet.java

```
package com.example;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

import java.io.IOException;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;

@WebServlet("/EmployeeServlet")
public class EmployeeServlet extends HttpServlet {

    private static final String URL = "jdbc:mysql://localhost:3306/javadb";
    private static final String USER = "root";
```

```
private static final String PASSWORD = "1234";

private Connection getConnection() throws SQLException {
   return DriverManager.getConnection(URL, USER, PASSWORD);
}

@Override
public void init() throws ServletException {
   try {
      Class.forName("com.mysql.cj.jdbc.Driver");
   } catch (ClassNotFoundException e) {
      throw new ServletException(e);
   }
}

@Override
protected void doGet(HttpServletRequest request,
                HttpServletResponse response)
      throws ServletException, IOException {
   processRequest(request, response);
}

@Override
protected void doPost(HttpServletRequest request,
                 HttpServletResponse response)
      throws ServletException, IOException {
   processRequest(request, response);
}

private void processRequest(HttpServletRequest request,
                       HttpServletResponse response)
      throws ServletException, IOException {

   response.setContentType("text/html;charset=UTF-8");
   request.setCharacterEncoding("UTF-8");

   String action = request.getParameter("action");

   try (Connection con = getConnection()) {

      if ("add".equals(action)) {
         String sql = "INSERT INTO employee (id, name, department, salary) VALUES (?, ?, ?, ?)";

         try (PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setInt(1, Integer.parseInt(request.getParameter("id")));
            ps.setString(2, request.getParameter("name"));
            ps.setString(3, request.getParameter("department"));
            ps.setDouble(4, Double.parseDouble(request.getParameter("salary")));
```

```
            int rows = ps.executeUpdate();
            request.setAttribute("message",
                    rows > 0 ? "Employee added successfully!" : "Employee could not be added.");
        }

    } else if ("update".equals(action)) {
        String sql = "UPDATE employee SET name=?, department=?, salary=? WHERE id=?";

        try (PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setString(1, request.getParameter("name"));
            ps.setString(2, request.getParameter("department"));
            ps.setDouble(3, Double.parseDouble(request.getParameter("salary")));
            ps.setInt(4, Integer.parseInt(request.getParameter("id")));

            int rows = ps.executeUpdate();
            request.setAttribute("message",
                    rows > 0 ? "Employee updated successfully!" : "Employee not found.");
        }

    } else if ("delete".equals(action)) {
        String sql = "DELETE FROM employee WHERE id=?";

        try (PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setInt(1, Integer.parseInt(request.getParameter("id")));

            int rows = ps.executeUpdate();
            request.setAttribute("message",
                    rows > 0 ? "Employee deleted successfully!" : "Employee not found.");
        }

    } else if ("search".equals(action)) {
        String sql = "SELECT * FROM employee WHERE id=?";

        try (PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setInt(1, Integer.parseInt(request.getParameter("id")));

            try (ResultSet rs = ps.executeQuery()) {
                if (rs.next()) {
                    Employee employee = new Employee(
                            rs.getInt("id"),
                            rs.getString("name"),
                            rs.getString("department"),
                            rs.getDouble("salary")
                    );
                    request.setAttribute("employee", employee);
                } else {
                    request.setAttribute("message", "Employee not found.");
```

```
                }
            }
        }

    } else if ("display".equals(action)) {
        List<Employee> employees = new ArrayList<>();
        String sql = "SELECT * FROM employee ORDER BY id";

        try (PreparedStatement ps = con.prepareStatement(sql);
             ResultSet rs = ps.executeQuery()) {

            while (rs.next()) {
                employees.add(new Employee(
                        rs.getInt("id"),
                        rs.getString("name"),
                        rs.getString("department"),
                        rs.getDouble("salary")
                ));
            }
        }

        request.setAttribute("employees", employees);
    }

} catch (NumberFormatException e) {
    request.setAttribute("message", "Please enter valid numeric values.");
} catch (SQLException e) {
    request.setAttribute("message", "Database error: " + e.getMessage());
}

request.getRequestDispatcher("employee.jsp").forward(request, response);
}
}
```

32. Develop a Product Management System using JSP, JSTL and EL. Retrieve product records from a database and display them using <c:forEach> and EL.

products.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ page import="java.sql.*" %>
<%@ page import="java.util.*" %>
<%@ page import="com.example.Product" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>
```

```
<%
   List<Product> products = new ArrayList<>();

   try {
      Class.forName("com.mysql.cj.jdbc.Driver");

      Connection con = DriverManager.getConnection(
         "jdbc:mysql://localhost:3306/javadb",
         "root",
         "1234"
      );

      PreparedStatement ps = con.prepareStatement("SELECT * FROM product");
      ResultSet rs = ps.executeQuery();

      while (rs.next()) {
         products.add(new Product(
            rs.getInt("id"),
            rs.getString("name"),
            rs.getInt("quantity"),
            rs.getDouble("price")
         ));
      }

      rs.close();
      ps.close();
      con.close();

   } catch (Exception e) {
      request.setAttribute("error", e.getMessage());
   }

   request.setAttribute("products", products);
%>

<!DOCTYPE html>
<html>
<head>
   <title>Product Management System</title>
</head>
<body>
   <h2>Product Management System</h2>
   <p>Name: Dhruv | Roll No: 01 | MCA_01</p>

   <c:if test="${not empty error}">
      <p><c:out value="${error}"/></p>
   </c:if>
```

```
    <table border="1" cellpadding="10">
        <tr>
            <th>Product ID</th>
            <th>Product Name</th>
            <th>Quantity</th>
            <th>Price</th>
            <th>Total Price</th>
        </tr>

        <c:forEach var="p" items="${products}">
            <tr>
                <td><c:out value="${p.id}"/></td>
                <td><c:out value="${p.name}"/></td>
                <td><c:out value="${p.quantity}"/></td>
                <td><c:out value="${p.price}"/></td>
                <td>${p.quantity * p.price}</td>
            </tr>
        </c:forEach>
    </table>
</body>
</html>
```

Product.java

```
package com.example;

public class Product {
    private int id;
    private String name;
    private int quantity;
    private double price;

    public Product(int id, String name, int quantity, double price) {
        this.id = id;
        this.name = name;
        this.quantity = quantity;
        this.price = price;
    }

    public int getId() {
        return id;
    }

    public String getName() {
        return name;
    }
```

```
    public int getQuantity() {
        return quantity;
    }

    public double getPrice() {
        return price;
    }
}
```

33. Create a JSP-based Login and Logout System using JSTL and EL. Use session scope to maintain the logged-in user and prevent access to the dashboard after logout.

login.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<!DOCTYPE html>
<html>
<head>
    <title>Login</title>
</head>
<body>
    <h2>Login Page</h2>
    <p>Name: Dhruv | Roll No: 01 | MCA_01</p>

    <c:if test="${not empty param.error}">
        <p>Invalid username or password!</p>
    </c:if>

    <form action="validate.jsp" method="post">
        Username:
        <input type="text" name="username" required>
        <br><br>

        Password:
        <input type="password" name="password" required>
        <br><br>

        <input type="submit" value="Login">
    </form>
</body>
</html>
```

validate.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:choose>
   <c:when test="${param.username == 'Dhruv' and param.password == '1234'}">
      <c:set var="loggedInUser" value="${param.username}" scope="session"/>
      <c:redirect url="dashboard.jsp"/>
   </c:when>

   <c:otherwise>
      <c:redirect url="login.jsp?error=1"/>
   </c:otherwise>
</c:choose>
```

dashboard.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:choose>
   <c:when test="${empty sessionScope.loggedInUser}">
      <c:redirect url="login.jsp"/>
   </c:when>

   <c:otherwise>
      <!DOCTYPE html>
      <html>
      <head>
         <title>Dashboard</title>
      </head>
      <body>
         <h2>Welcome, <c:out value="${sessionScope.loggedInUser}"/>!</h2>
         <p>You are successfully logged in.</p>

         <p>Name: Dhruv | Roll No: 01 | MCA_01</p>

         <form action="logout.jsp" method="post">
            <input type="submit" value="Logout">
         </form>
      </body>
      </html>
   </c:otherwise>
```

```
</c:choose>
```

logout.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:remove var="loggedInUser" scope="session"/>

<%
   session.invalidate();
   response.setHeader("Cache-Control", "no-cache, no-store, must-revalidate");
   response.setHeader("Pragma", "no-cache");
   response.setDateHeader("Expires", 0);
%>

<c:redirect url="login.jsp"/>
```

34. Develop a JSP application that retrieves records from a database and uses JSTL <c:forEach>, <c:if> and EL to display records conditionally.

students.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ page import="java.sql.*" %>
<%@ page import="java.util.*" %>
<%@ page import="com.example.Student" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<%
   List<Student> students = new ArrayList<>();

   try {
      Class.forName("com.mysql.cj.jdbc.Driver");

      Connection con = DriverManager.getConnection(
         "jdbc:mysql://localhost:3306/javadb",
         "root",
         "1234"
      );

      PreparedStatement ps = con.prepareStatement("SELECT * FROM student");
      ResultSet rs = ps.executeQuery();
```

```
        while (rs.next()) {
            students.add(new Student(
                rs.getInt("id"),
                rs.getString("name"),
                rs.getString("course"),
                rs.getInt("semester"),
                rs.getDouble("marks")
            ));
        }

        rs.close();
        ps.close();
        con.close();

    } catch (Exception e) {
        request.setAttribute("error", e.getMessage());
    }

    request.setAttribute("students", students);
%>

<!DOCTYPE html>
<html>
<head>
    <title>Student Records</title>
</head>
<body>
    <h2>Student Records</h2>
    <p>Name: Dhruv | Roll No: 01 | MCA_01</p>

    <c:if test="${not empty error}">
        <p>Database Error: <c:out value="${error}"/></p>
    </c:if>

    <c:if test="${empty students and empty error}">
        <p>No student records found.</p>
    </c:if>

    <c:if test="${not empty students}">
        <table border="1" cellpadding="10">
            <tr>
                <th>ID</th>
                <th>Name</th>
                <th>Course</th>
                <th>Semester</th>
                <th>Marks</th>
                <th>Result</th>
            </tr>
```

```
        <c:forEach var="s" items="${students}">
          <tr>
            <td><c:out value="${s.id}"/></td>
            <td><c:out value="${s.name}"/></td>
            <td><c:out value="${s.course}"/></td>
            <td><c:out value="${s.semester}"/></td>
            <td><c:out value="${s.marks}"/></td>

            <td>
              <c:if test="${s.marks >= 40}">
                Pass
              </c:if>
              <c:if test="${s.marks < 40}">
                Fail
              </c:if>
            </td>
          </tr>
        </c:forEach>
      </table>
    </c:if>
</body>
</html>
```

Student.java

```
package com.example;

public class Student {
    private int id;
    private String name;
    private String course;
    private int semester;
    private double marks;

    public Student(int id, String name, String course, int semester, double marks) {
        this.id = id;
        this.name = name;
        this.course = course;
        this.semester = semester;
        this.marks = marks;
    }

    public int getId() {
        return id;
    }
```

```
    public String getName() {
        return name;
    }

    public String getCourse() {
        return course;
    }

    public int getSemester() {
        return semester;
    }

    public double getMarks() {
        return marks;
    }
}
```

35. Develop a College Student Management System using JSP, JSTL, EL and JDBC. The system should provide:

Student Registration
Student Login
Student List
Search Student
Update Student
Delete Student
Session Management
Logout

Student.java

```
package com.example;

public class Student {
    private int id;
    private String name;
    private String email;
    private String course;
    private String password;

    public Student() {}

    public Student(int id, String name, String email, String course, String password) {
        this.id = id;
```

```
        this.name = name;
        this.email = email;
        this.course = course;
        this.password = password;
    }

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    public String getEmail() { return email; }
    public void setEmail(String email) { this.email = email; }

    public String getCourse() { return course; }
    public void setCourse(String course) { this.course = course; }

    public String getPassword() { return password; }
    public void setPassword(String password) { this.password = password; }
}
```

DBConnection.java

```
package com.example;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConnection {
    private static final String URL = "jdbc:mysql://localhost:3306/javadb";
    private static final String USER = "root";
    private static final String PASSWORD = "1234";

    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }
}
```

register.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<!DOCTYPE html>
```

```
<html>
<head><title>Student Registration</title></head>
<body>
  <h2>College Student Registration</h2>
  <p>Name: Dhruv | Roll No: 01 | MCA_01</p>

  <c:if test="${not empty param.error}">
    <p>Registration failed. Email may already exist.</p>
  </c:if>

  <form action="StudentServlet" method="post">
    <input type="hidden" name="action" value="register">

    Student ID: <input type="number" name="id" required><br><br>
    Name: <input type="text" name="name" required><br><br>
    Email: <input type="email" name="email" required><br><br>
    Course: <input type="text" name="course" value="MCA_01" required><br><br>
    Password: <input type="password" name="password" required><br><br>

    <input type="submit" value="Register">
  </form>

  <a href="login.jsp">Login</a>
</body>
</html>
```

login.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<!DOCTYPE html>
<html>
<head><title>Student Login</title></head>
<body>
  <h2>Student Login</h2>

  <c:if test="${not empty param.error}">
    <p>Invalid email or password.</p>
  </c:if>

  <form action="StudentServlet" method="post">
    <input type="hidden" name="action" value="login">

    Email: <input type="email" name="email" required><br><br>
    Password: <input type="password" name="password" required><br><br>
```

```
        <input type="submit" value="Login">
    </form>

    <a href="register.jsp">Register New Student</a>
</body>
</html>
```

dashboard.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:choose>
    <c:when test="${empty sessionScope.studentId}">
        <c:redirect url="login.jsp"/>
    </c:when>
    <c:otherwise>
        <!DOCTYPE html>
        <html>
        <head><title>Student Dashboard</title></head>
        <body>
            <h2>Welcome, <c:out value="${sessionScope.studentName}"/>!</h2>

            <p>Student ID: <c:out value="${sessionScope.studentId}"/></p>
            <p>Email: <c:out value="${sessionScope.studentEmail}"/></p>
            <p>Course: <c:out value="${sessionScope.studentCourse}"/></p>

            <a href="StudentServlet?action=list">Student List</a><br><br>
            <a href="search.jsp">Search Student</a><br><br>
            <a href="logout.jsp">Logout</a>
        </body>
        </html>
    </c:otherwise>
</c:choose>
```

search.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:if test="${empty sessionScope.studentId}">
    <c:redirect url="login.jsp"/>
</c:if>

<!DOCTYPE html>
```

```
<html>
<head><title>Search Student</title></head>
<body>
  <h2>Search Student</h2>

  <form action="StudentServlet" method="get">
    <input type="hidden" name="action" value="search">
    Enter Student ID:
    <input type="number" name="id" required>
    <input type="submit" value="Search">
  </form>

  <c:if test="${not empty param.error}">
    <p>Student not found.</p>
  </c:if>

  <a href="dashboard.jsp">Back to Dashboard</a>
</body>
</html>
```

students.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:if test="${empty sessionScope.studentId}">
  <c:redirect url="login.jsp"/>
</c:if>

<!DOCTYPE html>
<html>
<head><title>Student List</title></head>
<body>
  <h2>College Student List</h2>

  <c:if test="${empty students}">
    <p>No students found.</p>
  </c:if>

  <c:if test="${not empty students}">
    <table border="1" cellpadding="8">
      <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Email</th>
        <th>Course</th>
        <th>Actions</th>
```

```
        </tr>

        <c:forEach var="s" items="${students}">
          <tr>
            <td><c:out value="${s.id}"/></td>
            <td><c:out value="${s.name}"/></td>
            <td><c:out value="${s.email}"/></td>
            <td><c:out value="${s.course}"/></td>
            <td>
              <a
href="edit.jsp?id=s.id&name =${s.name}&email=s.email&course =${s.course}">Edit</a>
              |
              <form action="StudentServlet" method="post" style="display:inline">
                <input type="hidden" name="action" value="delete">
                <input type="hidden" name="id" value="${s.id}">
                <input type="submit" value="Delete"
                       onclick="return confirm('Delete this student?')">
              </form>
            </td>
          </tr>
        </c:forEach>
      </table>
    </c:if>

    <br><a href="dashboard.jsp">Back to Dashboard</a>
</body>
</html>
```

edit.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:if test="${empty sessionScope.studentId}">
    <c:redirect url="login.jsp"/>
</c:if>

<!DOCTYPE html>
<html>
<head><title>Update Student</title></head>
<body>
    <h2>Update Student</h2>

    <form action="StudentServlet" method="post">
        <input type="hidden" name="action" value="update">

        Student ID:
```

```
        <input type="number" name="id" value="<c:out value='${param.id}'/>" readonly><br><br>

        Name:
        <input type="text" name="name" value="<c:out value='${param.name}'/>" required><br><br>

        Email:
        <input type="email" name="email" value="<c:out value='${param.email}'/>"
required><br><br>

        Course:
        <input type="text" name="course" value="<c:out value='${param.course}'/>"
required><br><br>

        <input type="submit" value="Update Student">
    </form>
</body>
</html>
```

logout.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<%
    session.invalidate();
    response.setHeader("Cache-Control", "no-cache, no-store, must-revalidate");
    response.setHeader("Pragma", "no-cache");
    response.setDateHeader("Expires", 0);
%>

<c:redirect url="login.jsp"/>
```

StudentServlet.java

```
package com.example;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

import java.io.IOException;
import java.sql.*;
import java.util.ArrayList;
```

```
import java.util.List;

@WebServlet("/StudentServlet")
public class StudentServlet extends HttpServlet {

    @Override
    public void init() throws ServletException {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            throw new ServletException(e);
        }
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    private void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        request.setCharacterEncoding("UTF-8");
        response.setContentType("text/html;charset=UTF-8");

        String action = request.getParameter("action");

        try (Connection con = DBConnection.getConnection()) {

            if ("register".equals(action)) {
                String sql = "INSERT INTO student (id, name, email, course, password) VALUES (?, ?, ?,
?, ?)";

                try (PreparedStatement ps = con.prepareStatement(sql)) {
                    ps.setInt(1, Integer.parseInt(request.getParameter("id")));
                    ps.setString(2, request.getParameter("name"));
                    ps.setString(3, request.getParameter("email"));
                    ps.setString(4, request.getParameter("course"));
                    ps.setString(5, request.getParameter("password"));
                    ps.executeUpdate();
                }
```

```
    response.sendRedirect("login.jsp");
    return;

} else if ("login".equals(action)) {
    String sql = "SELECT * FROM student WHERE email=? AND password=?";

    try (PreparedStatement ps = con.prepareStatement(sql)) {
        ps.setString(1, request.getParameter("email"));
        ps.setString(2, request.getParameter("password"));

        try (ResultSet rs = ps.executeQuery()) {
            if (rs.next()) {
                HttpSession session = request.getSession();
                session.setAttribute("studentId", rs.getInt("id"));
                session.setAttribute("studentName", rs.getString("name"));
                session.setAttribute("studentEmail", rs.getString("email"));
                session.setAttribute("studentCourse", rs.getString("course"));

                response.sendRedirect("dashboard.jsp");
            } else {
                response.sendRedirect("login.jsp?error=1");
            }
        }
    }
    return;

} else {
    HttpSession session = request.getSession(false);
    if (session == null || session.getAttribute("studentId") == null) {
        response.sendRedirect("login.jsp");
        return;
    }
}

if ("list".equals(action)) {
    List<Student> students = new ArrayList<>();

    try (PreparedStatement ps =
             con.prepareStatement("SELECT id, name, email, course FROM student");
         ResultSet rs = ps.executeQuery()) {

        while (rs.next()) {
            students.add(new Student(
                    rs.getInt("id"),
                    rs.getString("name"),
                    rs.getString("email"),
                    rs.getString("course"),
```

```
                ""
            ));
        }
    }

    request.setAttribute("students", students);
    request.getRequestDispatcher("students.jsp").forward(request, response);
    return;
}

if ("search".equals(action)) {
    try (PreparedStatement ps = con.prepareStatement(
            "SELECT id, name, email, course FROM student WHERE id=?")) {

        ps.setInt(1, Integer.parseInt(request.getParameter("id")));

        try (ResultSet rs = ps.executeQuery()) {
            if (rs.next()) {
                Student s = new Student(
                        rs.getInt("id"),
                        rs.getString("name"),
                        rs.getString("email"),
                        rs.getString("course"),
                        ""
                );
                request.setAttribute("students", List.of(s));
                request.getRequestDispatcher("students.jsp").forward(request, response);
            } else {
                response.sendRedirect("search.jsp?error=1");
            }
        }
    }
    return;
}

if ("update".equals(action)) {
    String sql = "UPDATE student SET name=?, email=?, course=? WHERE id=?";

    try (PreparedStatement ps = con.prepareStatement(sql)) {
        ps.setString(1, request.getParameter("name"));
        ps.setString(2, request.getParameter("email"));
        ps.setString(3, request.getParameter("course"));
        ps.setInt(4, Integer.parseInt(request.getParameter("id")));
        ps.executeUpdate();
    }

    response.sendRedirect("StudentServlet?action=list");
    return;
```

```
            }

            if ("delete".equals(action)) {
                try (PreparedStatement ps =
                        con.prepareStatement("DELETE FROM student WHERE id=?")) {
                    ps.setInt(1, Integer.parseInt(request.getParameter("id")));
                    ps.executeUpdate();
                }

                response.sendRedirect("StudentServlet?action=list");
                return;
            }

            response.sendRedirect("dashboard.jsp");

        } catch (SQLException | NumberFormatException e) {
            throw new ServletException("Student management operation failed.", e);
        }
    }
}
```